Final Report

Email, secure mail, its threats, detection

Principles of information security

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OBSAH

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Introduction to the issue and specification of the research area

Description of email

Electronic mail, or "email," is a digital communication method that uses electronic devices to send and deliver messages across computer networks. The term "email" refers both to the system by which messages are transmitted and to the individual messages themselves. Email began in the 1970s, with Ray Tomlinson creating a method to send messages between computer systems on ARPANET. Over time, email evolved with email client software (e.g. Outlook) and web browser-based interface (e.g. Gmail) to make sending and receiving messages via the Internet easier. [1] [2]

Underlying principles of email

Email depends on mail servers: Outgoing servers to forward messages, and incoming servers to store and deliver them to recipients. Outgoing servers use SMTP (Simple Mail Transfer Protocol) to send emails, while incoming servers often use IMAP or POP3 for retrieving messages. [3] DNS (Domain Name System), especially MX (Mail Exchange) records, plays a critical role in routing email to the correct destination servers. An email "envelope" carries metadata like sender and recipient addresses that SMTP relies on for delivery. Because email travels through intermediate hops, delays, misrouting, or loss can occur due to network or configuration issues. To counter abuse (spam, phishing, spoofing), email systems adopt authentication standards (like SPF, DKIM, DMARC) and filtering mechanisms to validate message integrity and trust. [1] [2]

Email Structure

Original RFC822 (from year 1982) - Standard for the format of ARPA internet text messages - states that a message consists of header fields and a body, which is optional part of an email. Body is described as a sequence of ASCII characters, separated from the header by a CRLF. [4]

The header fields are described as a line of text consisting of ASCII characters containing field-name and field-body. The sequence of characters does not to be in a single

line, when it is split into multi lines it is called “folding”, opposite process of turning multi line text into a single line sequence of characters is called “unfolding”. [4]

Header field syntax is as following:

field = field-name “:” [field-body] CRLF

field-name = 1*<any CHAR, excluding CTLs, SPACE, and “:”>

field-body = field-body-contents

field-body-contents = <ASCII chars making up the field-body> [4]

RFC822 also states the syntax of an email as following: Header fields do not require to be in a specific order, the only rule is that the body has to be behind the header fields. It is recommended that the header fields are in this order:

1. **Return-Path**: Contains information about the address and route back to the sender. This field is added by the final transport of message to its recipients.
2. **Received**: Copy of this field I added by each transport service that relays the message.
3. **Date**: Syntax: [day “,”] date time ; dd mm yy ; hh:mm:ss zzz.
4. **From**: Contains the information about the person who sent the message. It is not possible to fill the body-field of From with list of senders, only 1 address can be present in this field. If this field is empty, then the field “Sender” must be present and filled.
5. **Subject**: Summary or subject of the message.
6. **Sender**: This field is filled with authenticated identity of a person/system that sent the message. The idea behind is that this field substitutes the “From” field. This field has to be present if the field-body is not the same as “From”.
7. **To**: Contains the identity of the primary recipients.
8. **Cc**: Contains the identity of the secondary recipients (“cc” stands for “carbon copy”).
9. **Bcc**: Has the identity of additional recipients. Some systems only show this field in the author’s copy, not the primary or secondary recipients,

although they can respond also to these “hidden” recipients (“bcc” stands for “blind carbon copy”).

10. **Reply-To**: The content of this field-body is an address to which responses are to be sent. [4]

Body of an email is a sequence of US-ASCII characters. There are just 2 rules:

- CRLF has to be together, no CR or LF independently inside the message
- Lines of characters in the body are limited to 998 characters excluding CRLF [4]

Address is a key component of an email, it appears in various message header fields to identify sender and recipients of a message. An address could represent a single mailbox or a group of mailboxes. A group is useful when you are sending emails to more mailboxes. [4]

Mailbox can be either name-address or addr-spec. The name-address is in the form:

[display-name] “<” addr-spec “>”

An example of a name-address is: Nikola Sasinkova <xsasinkova@stuba.sk>

Whereas an addr-spec is just: xsasinkova@stuba.sk

A group is defined in a format:

display-name “:” [mailbox -list] “;”

mailbox-list = (mailbox *(“,” mailbox))

Addr-spec is an identifier that satisfies these rules:

- addr-spec = local-part “@” domain
    - local-part = quoted-string
    - domain = domain-literal
        - domain-literal = “[“ (dtext / quoted-pair) “]”
            - dtext = CHARS excluding “[]\" and CR

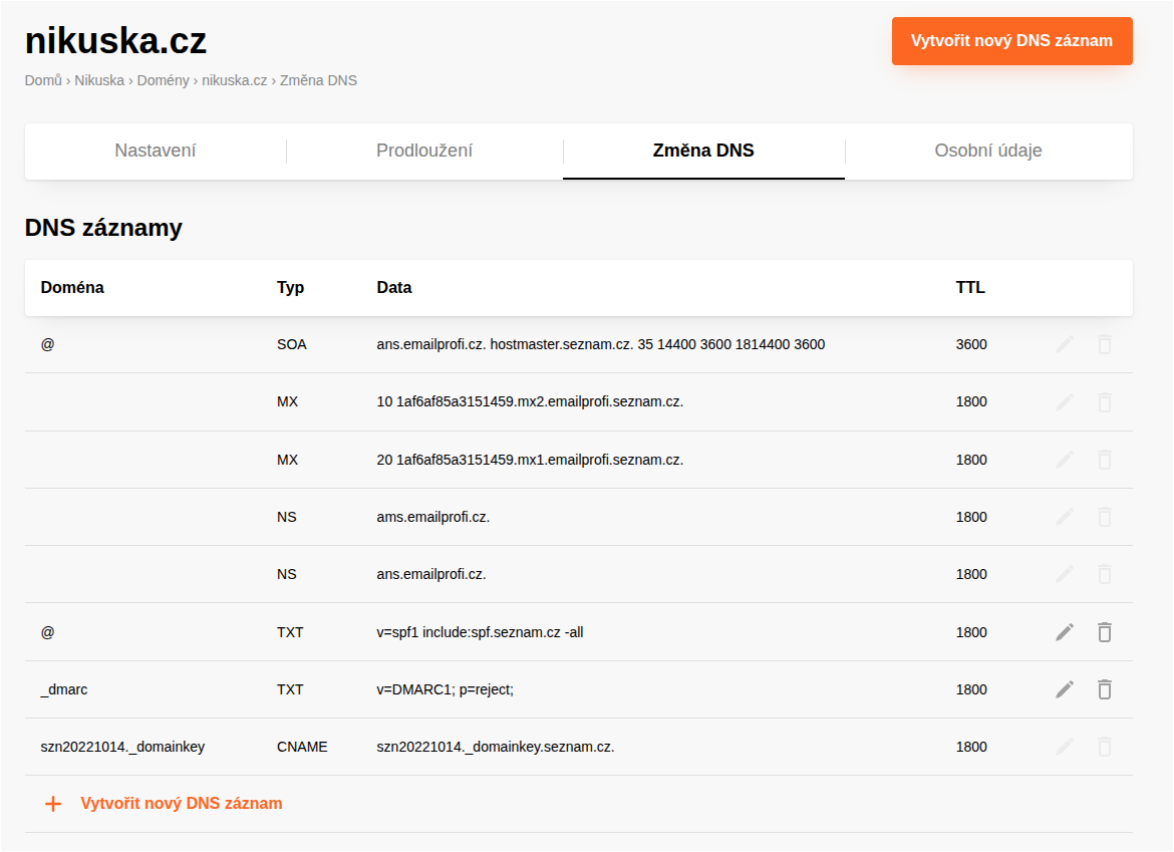
- Quoted-pair = "\" CHAR [4]

An example is: nikolka.sasinkova@gmail.com, where *nikolka.sasinkova* is local-part and *gmail.com* is a domain; or nikinka@nikuska.cz, where *nikinka* is local-part and *nikuska.cz* is a domain. [4]

Forwarding of an email happens by adding or modifying some header fields to the original header, e.g. "Resent-From" indicates the person who sent this specific email, whereas the field "From" contains the information of the original email sender. [4]

Emailprofi setup

First, I created an account at seznam.cz which I used to register my domain nikuska.cz. After registering I can access and change DNS records such as SPF and DMARC. [5]



Doména	Typ	Data	TTL
@	SOA	ans.emailprofi.cz. hostmaster.seznam.cz. 35 14400 3600 1814400 3600	3600
	MX	10 1af6af85a3151459.mx2.emailprofi.seznam.cz.	1800
	MX	20 1af6af85a3151459.mx1.emailprofi.seznam.cz.	1800
	NS	ams.emailprofi.cz.	1800
	NS	ans.emailprofi.cz.	1800
@	TXT	v=spf1 include:spf.seznam.cz -all	1800
_dmarc	TXT	v=DMARC1; p=reject;	1800
szn20221014._domainkey	CNAME	szn20221014._domainkey.seznam.cz.	1800

Sender Policy Framework

SPF (Sender Policy Framework) record is a type of DNS TXT record that specifies which mail servers are allowed to send emails on behalf of a domain. A TXT (text) record in DNS allows domain administrators to store plain text information in the domain's DNS settings. DNS records are a set of instructions stored on DNS servers that map human-readable domain names (like www.google.com) to machine-readable IP addresses (like 172.217.160.142). Originally, these records were used to post general information about a domain, but over time, they have been adapted for additional uses. [6]

SPF was introduced because the SMTP protocol, which handles email delivery, does not verify the sender's "from" address by default. SPF is used for identifying whether the email comes from a trusted source. Without SPF or similar email authentication methods, attackers can easily fake sender addresses, making it easier to deceive recipients into responding or giving up sensitive information. [6]

Checking an SPF record is quite simple, the mail server that receives an email from mail server that sends the message will check the return-path domain and search for its SPF record. If the check is successful, it then searches for the sender mail server IP address in its list of authorized senders. If both checks are successful the email will go through, otherwise the email will be rejected or marked as spam. [6]

Example of SPF record from EmailPprofi:

@ TXT v=spf1 include:spf.seznam.cz ~all 1800

v=spf1 - Tells the server that this is content of SPF record.

include:... - This tag instructs the server to find out that third-party organizations are authorized to send emails with this specific domain (seznam.cz in my case). This domain is then considered authorized.

~all - This flag tells the server that addresses that are not listed in SPF record will be marked as spam but will be accepted. [6]

There are also 2 other “all” flags:

-all - Addresses that are not listed in SPF record are not authorized to send emails and therefore should be rejected.

+all - Tells the server that any server can send email on behalf of the domain. [6]

There are some rules when configurating SPF:

- Only 1 SPF record per domain
- Record must end with “all” flag or include “redirect” component
- Record cannot contain uppercase characters [6]

DomainKeys Identified Mail

DomainKeys Identified Mail or shortly DKIM is an authentication method that prevents attackers from impersonating a verified or legitimate domain - sending an email from a domain that does not correspond to the sender. [7]

DKIM consists of 2 parts: The DKIM record and the DKIM header. The record is stored in the DNS records for the domain whereas the header is attached to all emails from the domain. [7]

DKIM uses digital signature scheme based on asymmetric cryptography, it uses public key to authenticate the server that sent the mail from corresponding domain, the public key is also used for verification of signatures. The private key is used by sender for signing the messages. The keys are generated by the email provider, the public key is given to the domain owner that is stored in the DKIM record. [7]

DKIM record is a DNS TXT record which stored the DKIM public key. It can also be a CNAME record that points to the key instead. It is saved under a special name, it follows this format:

[selector]._domainkey.[domain]

- The Selector field is issued by the email service provider used by the domain
- The Domain field is the email domain name [7]

My DKIM setup at Email Profi is a CNAME DNS record:

szn20221014._domainkey CNAME szn20221014._domainkey.seznam.cz. 1800

szn20221014 - The Selector.

Since I have a CNAME DNS record the data field is filled with the pointer to the key.

DKIM header is attached to the email header, it contains the digital signature which is a hash of the email body. The DKIM check can determine whether the email body has been changed or not. [7]

Domain-based Message Authentication Reporting and Conformance

DMARC or Domain-based Message Authentication Reporting and Conformance is a TXT DNS record used for authenticating email messages. DMARC policy tells what should happen to an email after checking SPF and DKIM. It determines what happens with an email if one or both checks fail, it could:

- Be marked as spam
- Get blocked
- Be delivered [9]

This policy is written in commands, my DMARC policy setup at Email Profi:

_dmarc TXT v=DMARC1; p=reject; 1800

_dmarc – Subdomain.

TXT - Type of DNS record.

In the Data field there is a string: "**v=DMARC1; p=quarantine; adkim=r; aspf=r;**"

It is interpreted as following:

v=DMARC1 - Tells that the TXT record contains a DMARC policy.

p=quarantine - Tells email servers that they should "quarantine" emails if they fail DKIM and SPF since they could be potential spam. P-value could be also set to "none", which will allow all emails to go through server even if the SPF and DKIM fail. Or the p-value could be "reject" which instructs the mail servers to block failed emails. [9]

adkim=r - Tells that DKIM checks are "relaxed" since adkim is "r", it could be changed to "s" which means "strict", this part is optional.

aspf=r - Sets SPF check to relaxed" since aspf is set to "r", it could be changed to "s" which means "strict" - same as DKIM.

Other optional parts:

rua=mailto:example@example.com - is set when the administrators want to send mails that failed SPF and DKIM as reports to third-party service to better setup DMARC. [8]

Security & types of attacks / threats

With electronic communication come security vulnerabilities and techniques used by attackers. Among the most common are the following attacks:

1. Phishing

Phishing is a foundational form of online fraud where attackers impersonate trustworthy entities to gather credentials from recipients, download malware on their devices and authorize illegitimate transactions. Rather than exploiting software vulnerabilities, it targets the human element of cybersecurity. Most phishing attempts are mass-distributed and use tactics that scare the recipient such as:

- Account suspension notices
- Password reset prompts
- Delivery failures
- Financial alerts

These messages are engineered to provoke fast reactions before the recipient can evaluate legitimacy. They typically include deceptive links that appear to lead to familiar domains but instead redirect to fake login portals designed to gather credentials. [9]

Attackers cast a wide net, assuming that even if a small portion of recipients respond, it yields sufficient payoff. Automation enables attackers to spoof email addresses, use look-alike domains and bypass basic filters. [9]

There are a few types of phishing:

- Smishing - SMS phishing
- Vishing - Voice phishing
- Spear phishing - targeted phishing [9]

2. Spear phishing

Spear phishing advances the concept of phishing but replaces mass deception with precision manipulation. Instead of impersonating a generic institution, the attacker profiles a specific individual or organization and builds credibility before executing the attack. Public data sources like:

- LinkedIn titles - Get information about people and their work titles
- Corporate press releases - Get email addresses, names, contact information
- Conference speaker lists - Get email addresses, names of speakers
- Compromised email archives - Already existing database of compromised information

provide enough intelligence to craft a message that feels contextually appropriate. [9]

More sophisticated spear phishing campaigns unfold gradually: Attackers may begin with believable correspondence to gain trust before escalating to a high-value request. Some advanced spear phishing attacks do not start with an immediate request. Instead, the attacker may first send harmless or casual messages to build trust with the victim. Once the relationship feels normal, they slowly move toward a more serious request, such as asking for access or money. These types of attacks are very difficult for normal security filters to detect, because they do not follow common spam patterns. The real danger is not in the technical structure of the email, but in how believable and personal the message appears. [9]

3. Spoofing

Spoofing is a deception technique where an attacker disguises the origin of a message, call, or network packet to make it look like it comes from a trusted source. The core moves are concealment (hiding the true sender) and mimicry (copying attributes: Names, logos, addresses). The attacker's goal is to lower the target's guard so the victim will share credentials, open attachments, approve payments or click malicious links. Consequences range from credential theft and malware infection to financial fraud and reputational damage. [10]

Spoofing comes in several forms depending on the channel and technique:

- **Email spoofing** occurs when attackers forge the sender's address to make a message appear from a trusted source, often to steal credentials or deliver malware.
- **Caller ID spoofing** disguises the origin of a phone call, tricking victims into sharing sensitive information or authorizing transactions.
- **Website spoofing** involves creating a fake site that mimics a legitimate one to capture login credentials or financial information.
- **IP and network spoofing** falsify packet information, making it appear as if network traffic comes from a trusted device, often used in attacks like DoS.

Each type relies on exploiting trust and familiarity rather than breaking software directly. [10]

Detecting spoofing requires both technical monitoring and awareness of unusual patterns. Email spoofing can be identified through authentication checks like SPF, DKIM, and DMARC, which verify that messages are sent from authorized servers. Network spoofing may be detected with anomaly-based intrusion detection systems that flag unusual IP traffic or address inconsistencies. Users can also contribute to detection by noticing signs such as mismatched sender names and addresses, unexpected requests, poor grammar, or unusual urgency. Combining automated detection with human vigilance increases the likelihood of spotting spoofing attempts before they cause harm. [10]

Preventing spoofing involves a mix of technical controls, operational policies, and user training. Implementing strict already mentioned email authentication standards like SPF, DKIM, and DMARC reduces the risk of email impersonation. Multi-factor authentication protects accounts even if credentials are compromised, and secure network configurations reduce IP or packet spoofing threats. Operationally, organizations should verify financial and sensitive requests through secondary channels and educate employees on identifying suspicious messages or calls. Regular monitoring of brand impersonation and look-alike domains also helps organizations proactively block spoofing attacks. [10]

4. Denial-of-Service

A Denial-of-Service (DoS) attack is a malicious attempt to disrupt the normal operations of a targeted server, service or network by overwhelming it with an internet traffic. The primary goal is to exhaust the resources of the target system, rendering it unavailable to legitimate users. DoS attacks typically fall into two categories:

- Buffer overflow attack: Exploits vulnerabilities to consume system resources.
- Flood attacks: Saturate the target with excessive traffic.

This results in degraded performance or complete unavailability of the service. [11]

The history of DoS attacks dates back to the early days of the internet, with notable incidents highlighting their potential for disruption. Sophistication of these attacks has increased, leading to the development of advanced detection and prevention mechanisms. Detecting DoS attacks involves monitoring network traffic for unusual patterns, such as a sudden spike in requests or traffic from a single source. Intrusion detection systems and traffic analysis tools are commonly used to identify potential attacks. Preventing DoS attacks requires a multi-layered approach, including rate limiting, firewalls and content delivery networks to absorb and mitigate malicious traffic. Additionally, implementing redundancy and failover systems can help maintain service availability during an attack. Organizations must also stay informed about emerging attack vectors and continuously update their defense strategies to address evolving threats. [11]

5. Trojan horse

A Trojan horse is a type of malware that hides its malicious purpose by pretending to be a legitimate program or file. Unlike worms or viruses that self-replicate, Trojans rely on tricking a user into installing or running them. For example, by posing as a useful utility, a media file or a software update. Once executed, a Trojan can perform many harmful actions in the background: Open a remote access channel, steal credentials or files, install additional malware (ransomware, banking trojans, or bots) or alter system settings to maintain persistence. Because Trojans exploit trust and user action rather than system flaws directly, they are often

delivered through deceptive downloads, email attachments, malicious advertisements or compromised websites. [12]

The Trojan horse concept borrows its name from the ancient ruse (conceal a weapon inside something trusted) and has been central to malware campaigns since the early days of personal computing. Detecting Trojans requires both signature-based tools and behavior analysis: antivirus and endpoint detection and response can flag known samples, while heuristic/behavioral engines sandboxes identify suspicious actions such as unexpected network connections, unusual file changes or the creation of persistent startup entries. [12]

There are many types of Trojan horse:

- Backdoor Trojan: Allows an attacker to secretly access the infected device and control it remotely, enabling actions like stealing files or installing more malware.
- Banker Trojan: Designed to steal financial data such as online banking passwords, card information and payment details.
- Downloader Trojan: Focuses on downloading and installing additional malicious programs once it has gained access to the system.
- Exploit Trojan: Uses known software vulnerabilities to execute harmful code and gain control of the device.
- Fake Antivirus Trojan: Pretends to be security software, shows fake virus alerts, and demands payment to “fix” problems that do not exist.
- Rootkit Trojan: Hides itself and other malware from the operating system and security tools to avoid detection.
- Remote Access Trojan: Provides full remote control of the infected device, allowing attackers to monitor activity or take over system functions. [12]

Analysis of the selected tools for experimentation

1. Email Profi

Email Profi is an email hosting service operated by Seznam.cz that lets users run professional email addresses under their own custom domains while using the familiar Seznam webmail interface. The service supports an unlimited number of mailboxes and storage space and includes integrated tools such as a calendar and smart anti-spam filtering. You can either connect an existing domain by following these few steps:

1. Login to Email Profi with your @seznam account
2. Click on button "Add new domain"
3. State the name of your domain that you want to link with your account
4. Accept the license agreement and click on "I own a domain"
5. Generate your own new unique MX records, update the MX records in DNS settings
6. Create your mail inbox [14]

Or register a new domain directly through the platform by these steps:

1. Login to Email Profi your @seznam.cz account
2. If you do not own any domain, you will see a button "Add new domain"
3. Enter the name of your new domain, note: it can only be domain with ending .cz
4. Accept the license agreement, fill the information about the buyer and buy your new domain [13] [15]

Pros of using Email Profi:

- Unlimited users and storage - no strict limit on data usage and mailbox creation
- Users can work in the Seznam webmail environment they may already know
- Comes with a calendar, anti-spam filters and support for mobile and desktop clients [13]

Cons of using Email Profi:

- If you don’t have a domain, you must buy one
- You’re tied to Seznam’s infrastructure - if Seznam has downtime or policy changes, your email could be affected
- You must manage MX records and DNS correctly
- If SPF, DKIM, DMARC are not properly configured, outgoing emails may be flagged as spam or rejected [13]

2. Mail-Tester

Mail-tester is a custom tool of a few software developers to quickly test the quality of their newsletter. There is a web-interface and also paid API which you can use in your software to automate mail testing. The backend of the Mail-tester is a free open-source Apache SpamAssassin - a widely used spam filter. Since you can not ensure that your email is delivered to everybody correctly (it could be blocked or marked as spam), but you can do a lot to improve your chances by analyzing and configurating your email setup. Also, all spam filters work differently, so each mail provider will have a different way to mark an email as a spam. [16]

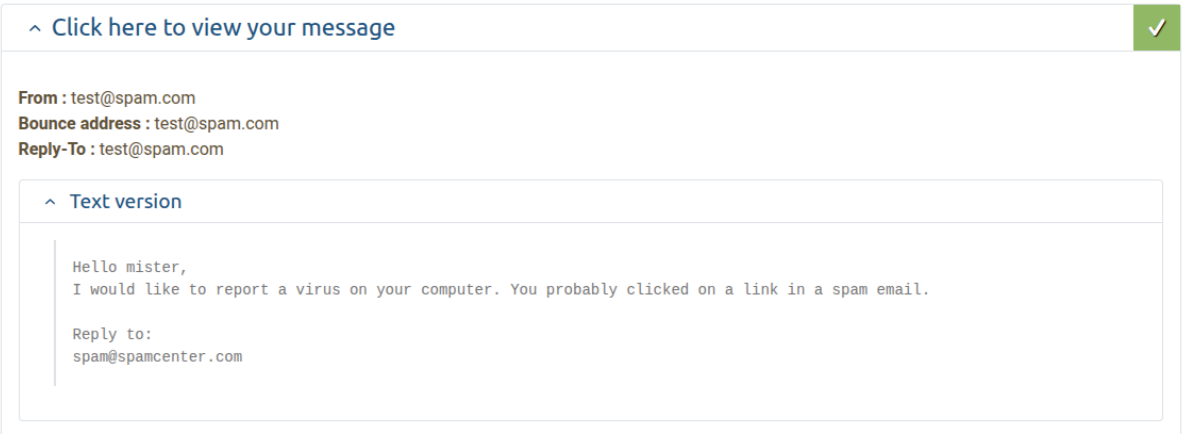
Mail testing workflow:

1. Mail-tester generates a random email address each time you access https://www.mail-tester.com/
2. You send a message with recipient sent to this address
3. Mail/tester assigns a score (up to 10) to your sent message
    - A positive score means good results
    - A negative score means the opposite
4. You can check the details of the evaluation (available for 7 days, 30 days if you are logged in) [16]

In Mail-tester web application after evaluating the score of your email you can analyze these attributes:

- Your message:
    - Text version
    - Source
- SpamAssassin score, score below -5 is considered spam
- Authentication:
    - SPF
    - DKIM
    - DMARC
    - Server association with website

    - Domain name association to a mail server
    - Hostname assignation to a server
- Message formation:
    - Validation of images
    - Content safety
    - Use of URL shortener system
    - Content of List-Unsubscribe header
- Check if your IP address is blocklisted [17]



Source

```
Received: by mail-tester.com (Postfix, from userid 500)
        id DDA8AA0C37; Sat, 18 Oct 2025 17:28:19 +0200 (CEST)
X-Spam-Checker-Version: SpamAssassin 3.4.2 (2018-09-13) on mail-tester.com
X-Spam-Level:
X-Spam-Status: No/-2.0/5.0
X-Spam-Test-Scores: RCVD_IN_DNSWL_BLOCKED=0.001,RCVD_IN_RP_SAFE=-2,
        SPF_HELO_PASS=-0.001,SPF_NONE=0.001,URIBL_BLOCKED=0.001
X-Spam-Last-External-IP: 114.29.236.247
X-Spam-Last-External-HELO: emkei.cz
X-Spam-Last-External-rDNS: emkei.cz
X-Spam-Date-of-Scan: Sat, 18 Oct 2025 17:28:19 +0200
X-Spam-Report:
        *  0.0 URIBL_BLOCKED ADMINISTRATOR NOTICE: The query to URIBL was
        *      blocked.  See
        *      http://wiki.apache.org/spamassassin/DnsBlocklists#dnsbl-block
        *      for more information.
        *      [URIs: spamcenter.com]
        *  0.0 RCVD_IN_DNSWL_BLOCKED RBL: ADMINISTRATOR NOTICE: The query to
        *      DNSWL was blocked.  See
        *      http://wiki.apache.org/spamassassin/DnsBlocklists#dnsbl-block
        *      for more information.
        *      [114.29.236.247 listed in list.dnswl.org]
        * -0.0 SPF_HELO_PASS SPF: HELO matches SPF record
        *  0.0 SPF_NONE SPF: sender does not publish an SPF Record
        * -2.0 RCVD_IN_RP_SAFE RBL: Sender in ReturnPath Safe - Contact
        *      safe-sa@returnpath.net
        *      [Excessive Number of Queries | <https://knowledge.validity.com/hc/en-us/articles/20961730681243>]
Received-SPF: Pass (sender SPF authorized) identity=helo; client-ip=114.29.236.247; helo=emkei.cz; envelope-from=test@sp
DMARC-Filter: OpenDMARC Filter v1.3.1 mail-tester.com ED5CFA0B90
Authentication-Results: mail-tester.com; dmarc=none header.from=spam.com
Received: from emkei.cz (emkei.cz [114.29.236.247])
        (using TLSv1.2 with cipher ADH-AES256-GCM-SHA384 (256/256 bits))
        (No client certificate requested)
        by mail-tester.com (Postfix) with ESMTPS id ED5CFA0B90
        for <test-qhnegyj1y@srv1.mail-tester.com>; Sat, 18 Oct 2025 17:28:17 +0200 (CEST)
Received: by emkei.cz (Postfix, from userid 33)
        id 94E8D3A22; Sat, 18 Oct 2025 17:28:14 +0200 (CEST)
To: test-qhnegyj1y@srv1.mail-tester.com
Subject: spam
From: test@spam.com
X-Priority: 3 (Normal)
Importance: Normal
Errors-To: test@spam.com
Reply-To: test@spam.com
Content-Type: text/plain; charset=utf-8
Message-Id: <20251018152814.94E8D3A22@emkei.cz>
Date: Sat, 18 Oct 2025 17:28:14 +0200 (CEST)

Hello mister,
I would like to report a virus on your computer. You probably clicked on a link in a spam email.

Reply to:
spam@spamcenter.com
```

SpamAssassin likes you

The famous spam filter SpamAssassin. Score: 2.
A score below -5 is considered spam.

2	RCVD_IN_RP_SAFE	Sender is in Return Path Safe (trusted relay)
0.001	SPF_HELO_PASS	SPF: HELO matches SPF record
-0.001	SPF_NONE	SPF: sender does not publish an SPF Record

You're not fully authenticated

-2

We check if the server you are sending from is authenticated

[SPF] spam.com does not allow your server 114.29.236.247 to use test@spam.com

-1

Sender Policy Framework (SPF) is an email validation system designed to prevent email spam by detecting email spoofing, a common vulnerability, by verifying sender IP addresses.

You do not have a SPF record, please add the following one to your domain **spam.com**:

```
v=spf1 a mx ip4:114.29.236.247 ~all
```

The Name Server handling the domain name **spam.com** is **ns-457.awsdns-57.com.**

Need guidance? View guides for each host.

Verification details:

```
dig +short TXT spam.com :
  • "facebook-domain-verification=lmipv3urvsjn478zk72282o6tveiwj"
  • "MS=ms11310971"
  • "google-site-verification=DcQxVy7tQU1BmbwgqoXugoWML2hqDP4wBTCi35Gl3QI"
  • "ahrefs-site-verification_0290cc135d85fa21dbead34bff9e781913a52f73a91e11ed63b7529744464778"

dig +short +time=3 +tries=1 TXT @ns-457.awsdns-57.com. spam.com :
  • "MS=ms11310971"
  • "ahrefs-site-verification_0290cc135d85fa21dbead34bff9e781913a52f73a91e11ed63b7529744464778"
  • "facebook-domain-verification=lmipv3urvsjn478zk72282o6tveiwj"
  • "google-site-verification=DcQxVy7tQU1BmbwgqoXugoWML2hqDP4wBTCi35Gl3QI"

spfquery --scope mfrom --id test@spam.com --ip 114.29.236.247 --helo-id emkei.cz :
  • none
  • spam.com: No applicable sender policy available
  • spam.com: No applicable sender policy available
  • Received-SPF: none (spam.com: No applicable sender policy available) receiver=ns303428.ip-94-23-206.eu; identity=ma
```

Your message is not signed with DKIM

-1

DomainKeys Identified Mail (DKIM) is a method for associating a domain name to an email message, thereby allowing a person, role, or organization to claim some responsibility for the message.

You do not have a DMARC record

A DMARC policy allows a sender to indicate that their emails are protected by SPF and/or DKIM, and give instruction if neither of those authentication methods passes. Please be sure you have a DKIM and SPF set before using DMARC.

You do not have a DMARC record, please add a **TXT** record to your domain **_dmarc.spam.com** with the following value:

```
v=DMARC1; p=reject
```

Verification details:

```
• mail-tester.com; dmarc=none header.from=spam.com
• From Domain: spam.com
• DKIM Domain:
```

Your server **114.29.236.247** is successfully associated with **emkei.cz**

Reverse DNS lookup or reverse DNS resolution (rDNS) is the determination of a domain name that is associated with a given IP address. Some companies such as AOL will reject any message sent from a server without rDNS, so you must ensure that you have one. You cannot associate more than one domain name with a single IP address.

```
Here are the tested values for this check:
• IP: 114.29.236.247
• HELO: emkei.cz
• rDNS: emkei.cz
```

Your domain name **spam.com** is assigned to a mail server.

*We check if there is a mail server (MX Record) behind your domain name **spam.com**.*

```
MX records (spam.com) :
  • 5 alt1.aspmx.l.google.com.
  • 1 aspmx.l.google.com.
  • 10 aspmx2.googlemail.com.
  • 10 aspmx4.googlemail.com.
  • 10 aspmx3.googlemail.com.
  • 5 alt2.aspmx.l.google.com.
  • 10 aspmx5.googlemail.com.
```

Your hostname **emkei.cz** is assigned to a server.

*We check if there is a server (A Record) behind your hostname **emkei.cz**.*

```
A records (emkei.cz) :
  • 114.29.236.247
```

You're not blocklisted

*Matches your server IP address (**114.29.236.247**) against 23 of the most common IPv4 blocklists.*

Not listed in Spamhaus SBL Advisory	Not listed in Spamhaus CSS Advisory	Not listed in Spamhaus XBL Advisory
Not listed in Spamhaus PBL Advisory	Not listed in Barracuda	Not listed in Hostkarma
Not listed in IMP-SPAM	Not listed in BACKSCATTERER	Not listed in China Anti-Spam Alliance
Not listed in LashBack	Not listed in mailspike	Not listed in REDHAWK
Not listed in SORBS (Relay)	Not listed in SORBS (last 48 hours)	Not listed in SORBS (last 28 days)
Not listed in SPAMCOP	Not listed in SEM-BACKSCATTER	Not listed in SEM-BLACK
Not listed in RATS-ALL	Not listed in PSBL	Not listed in SWINOG
Not listed in GBUdb Truncate	Not listed in Weighted Private Block List	

Your lovely total: 8/10

3. MXToolbox

MXToolbox is an online tool to verify email structure and check domain health. It can be used for checking DNS configurations, improving email deliverability, blacklist status and mail server availability. This multi-functioning tool can be used by both attackers and defenders. It is valuable for checking SPF, DKIM and DMARC records or verifying if third-party vendors are sending emails from blacklisted IPs. [18]

The main functions are:

- **DNS record lookup** - Validation of DNS including MX, A and TXT.
- **SMTP diagnostics** - Performs an open relay test - if SMTP server is incorrectly configurated and allows email to be forwarded by unauthorized parties.
- **Blacklist check** - Checks whether an email might be blocked or marked as spam based on scanning your IP address/domain.
- **Email header analysis** - Transforms an email header into human readable format to provide more informative data.
- **Monitoring and alerts** - Setup alerts when domain gets blacklisted. [18]

Pros of using MXToolbox:

- Offers wide range of tools for email and DNS analysis
- Provides real-time SMTP diagnosis [19]

Cons of using MXToolbox:

- System is too complex, so it is hard to navigate [19]

MXToolbox provides 3 subscription plans:

- Free - For 1 domain
- Delivery center - 129$/month - includes:
    - Inbox Placement Analysis
    - Recipient Complaint Reporting
    - Email Delivery Performance Reports
    - Email Configuration Analysis
    - Adaptive Sender Blacklist Monitoring
    - Inbound + Outbound MailFlow Monitoring
    - Domain Impersonation Protection
    - Domain(s): 5
    - Email Message Volume: 500,000
- Delivery center plus - 399$/month includes:
    - Inbox Placement Analysis
    - Recipient Complaint Reporting
    - Email Delivery Performance Reports
    - Email Configuration Analysis
    - Adaptive Sender Blacklist Monitoring, Real-time blacklist monitoring on all IP addresses that are actively sending mail
    - Inbound + Outbound MailFlow Monitoring
    - Domain Impersonation Protection
    - Advanced Email Delivery Threat Tools
    - SPF Flattening
    - Domain(s): 5
    - Email Message Volume: 5,000,000 [20]

4. Emkei's mailer

Emkei's mailer is a website dedicated for sending fake mails with various options including attachments, encryption and HTML editor. Its user interface is a website form with these fields:

- From name
- From e-mail
- To
- Subject
- Attachments
- Reply-to
- Errors-to
- Cc
- Bcc
- Priority
- X-Mailer
- Confirm delivery
- Confirm reading
- Add header
- SMTP Server
- Port
- Date
- Charset
- PGP/GPG encrypt
- Receiver's public key
- Content-type
- Text [21]

Experiment with Mail-tester

During this experiment I am going to compare a score of emails from Mail-tester with different setups. I set up DNS records on Email Profi from seznam.cz, I generated a tester email address on Mail-tester which I used for performing this experiment. For generating and sending fake emails I used online tool emkei.cz which is a website that lets you send a custom email that looks like spam. I tested various setups of DNS records and also various email bodies.

Mail-tester reports

Report from Mail-tester using personal email with valid text:

- Click here to view your message ✓
- SpamAssassin likes you ✓
- You're properly authenticated ✓

  We check if the server you are sending from is authenticated

  - [SPF] Your server **209.85.208.182** is authorized to use **nika.sasinkova@gmail.com** ✓
  - Your DKIM signature is valid ✓
  - Your message passed the DMARC test ✓
  - Your server **209.85.208.182** is successfully associated with **mail-lj1-f182.google.com** ✓
  - Your domain name **gmail.com** is assigned to a mail server. ✓
  - Your hostname **mail-lj1-f182.google.com** is assigned to a server. ✓
- Your message could be improved ✓
- You're not blocklisted ✓
- No broken links ✓

Your lovely total: 10/10

Report with DNS correct setup and valid text:

- Click here to view your message ✓
- SpamAssassin likes you ✓
- You're properly authenticated ✓

We check if the server you are sending from is authenticated

- [SPF] Your server **77.75.76.89** is authorized to use **nikinka@nikuska.cz** ✓
- Your DKIM signature is valid ✓
- Your message passed the DMARC test ✓
- Your server **77.75.76.89** is successfully associated with **mxb.seznam.cz** ✓
- Your domain name **nikuska.cz** is assigned to a mail server. ✓
- Your hostname **mxb.seznam.cz** is assigned to a server. ✓

- Your message could be improved ✓
- You're not blocklisted ✓
- No broken links ✓

Report using DNS incorrect setup with valid text:

- Click here to view your message ✓
- SpamAssassin likes you ✓
- You are not allowed to use one of your sender email addresses -3

  We check if the server you are sending from is authenticated

  - [SPF] nikuska.cz does not allow your server 77.75.76.89 to use nikinka@nikuska.cz -3
  - Your DKIM signature is valid ✓
  - Your message passed the DMARC test ✓
  - Your server **77.75.76.89** is successfully associated with **mxb.seznam.cz** ✓
  - Your domain name **nikuska.cz** is assigned to a mail server. ✓
  - Your hostname **mxb.seznam.cz** is assigned to a server. ✓
- Your message could be improved ✓
- You're not blocklisted ✓
- No broken links ✓

Your lovely total: 7/10

Fake email with valid text:

- Click here to view your message ✓
- SpamAssassin likes you ✓
- You're not fully authenticated -2

  We check if the server you are sending from is authenticated

  - [SPF] spam.com does not allow your server 114.29.236.247 to use test@spam.com -1
  - Your message is not signed with DKIM -1
  - You do not have a DMARC record ✓
  - Your server **114.29.236.247** is successfully associated with **emkei.cz** ✓
  - Your domain name **spam.com** is assigned to a mail server. ✓
  - Your hostname **emkei.cz** is assigned to a server. ✓
- Your message could be improved ✓
- You're not blocklisted ✓

Your lovely total: 8/10

Personal email with spam detected text:

Click here to view your message ✓

SpamAssassin does not like you -999.8

The famous spam filter SpamAssassin. Score: -999.8.
A score below -5 is considered spam.

-0.1	DKIM_SIGNED	Message has a DKIM or DK signature, not necessarily valid **This rule is automatically applied if your email contains a DKIM signature but other positive rules will also be added if your DKIM signature is valid. See immediately below.**
0.1	DKIM_VALID	Message has at least one valid DKIM or DK signature **Great! Your signature is valid**
0.1	DKIM_VALID_AU	Message has a valid DKIM or DK signature from author's domain **Great! Your signature is valid and it's coming from your domain name**
0.1	DKIM_VALID_EF	Message has a valid DKIM or DK signature from envelope-from domain
-0.001	FREEMAIL_FROM	Sender email is freemail **You're sending from a free email account**
-1000	GTUBE	Generic Test for Unsolicited Bulk Email
-0.001	HTML_MESSAGE	HTML included in message **No worry, that's expected if you send HTML emails**
-1.985	PYZOR_CHECK	Similar message reported on Pyzor (https://www.pyzor.org) https://pyzor.readthedocs.io/en/latest/ **Please test a real content, test Newsletters will always be flagged by Pyzor** **Adjust your message or request whitelisting (https://www.pyzor.org)**
0.001	RCVD_IN_MSPIKE_H2	Average reputation (+2) 209.85.167.43 listed in wl.mailspike.net
2	RCVD_IN_RP_SAFE	Sender is in Return Path Safe (trusted relay)
-0.001	SPF_HELO_NONE	SPF: HELO does not publish an SPF Record
0.001	SPF_PASS	SPF: sender matches SPF record **Great! Your SPF is valid**
-0.001	TVD_SPACE_RATIO	Uncommon space ratio

You're properly authenticated ✓

Your message could be improved ✓

You're not blocklisted ✓

No broken links ✓

Your lovely total: 0/10

DNS setup correct with spam detected text:

Click here to view your message

SpamAssassin does not like you -1000.5

The famous spam filter SpamAssassin. Score: -1000.5.
A score below -5 is considered spam.

-0.1	DKIM_SIGNED	Message has a DKIM or DK signature, not necessarily valid **This rule is automatically applied if your email contains a DKIM signature but other positive rules will also be added if your DKIM signature is valid. See immediately below.**
0.1	DKIM_VALID	Message has at least one valid DKIM or DK signature **Great! Your signature is valid**
0.1	DKIM_VALID_AU	Message has a valid DKIM or DK signature from author's domain **Great! Your signature is valid and it's coming from your domain name**
0.1	DKIM_VALID_EF	Message has a valid DKIM or DK signature from envelope-from domain
-1000	GTUBE	Generic Test for Unsolicited Bulk Email
-0.001	HTML_MESSAGE	HTML included in message **No worry, that's expected if you send HTML emails**
-0.724	MPART_ALT_DIFF	HTML and text parts are different **Make sure your TEXT version looks similar as the text inserted in your HTML version**
-1.985	PYZOR_CHECK	Similar message reported on Pyzor (https://www.pyzor.org) https://pyzor.readthedocs.io/en/latest/ **Please test a real content, test Newsletters will always be flagged by Pyzor** **Adjust your message or request whitelisting (https://www.pyzor.org)**
-0.001	RCVD_IN_MSPIKE_H4	Very Good reputation (+4) 77.75.78.89 listed in wl.mailspike.net
-0.001	RCVD_IN_MSPIKE_WL	Mailspike good senders
2	RCVD_IN_RP_SAFE	Sender is in Return Path Safe (trusted relay)
-0.001	SPF_HELO_NONE	SPF: HELO does not publish an SPF Record
0.001	SPF_PASS	SPF: sender matches SPF record **Great! Your SPF is valid**
-0.001	TVD_SPACE_RATIO	Uncommon space ratio

You're properly authenticated

Your message could be improved

You're not blocklisted

No broken links

Your lovely total: 0/10

DNS setup incorrect with spam detected text:

Click here to view your message

SpamAssassin does not like you -1000.5

The famous spam filter SpamAssassin. Score: -1000.5.
A score below -5 is considered spam.

-0.1	DKIM_SIGNED	Message has a DKIM or DK signature, not necessarily valid **This rule is automatically applied if your email contains a DKIM signature but other positive rules will also be added if your DKIM signature is valid. See immediately below.**
0.1	DKIM_VALID	Message has at least one valid DKIM or DK signature **Great! Your signature is valid**
0.1	DKIM_VALID_AU	Message has a valid DKIM or DK signature from author's domain **Great! Your signature is valid and it's coming from your domain name**
0.1	DKIM_VALID_EF	Message has a valid DKIM or DK signature from envelope-from domain
-1000	GTUBE	Generic Test for Unsolicited Bulk Email
-0.001	HTML_MESSAGE	HTML included in message **No worry, that's expected if you send HTML emails**
-0.724	MPART_ALT_DIFF	HTML and text parts are different **Make sure your TEXT version looks similar as the text inserted in your HTML version**
-1.985	PYZOR_CHECK	Similar message reported on Pyzor (https://www.pyzor.org) https://pyzor.readthedocs.io/en/latest/ **Please test a real content, test Newsletters will always be flagged by Pyzor** **Adjust your message or request whitelisting (https://www.pyzor.org)**
-0.001	RCVD_IN_MSPIKE_H4	Very Good reputation (+4) 77.75.78.89 listed in wl.mailspike.net
-0.001	RCVD_IN_MSPIKE_WL	Mailspike good senders
2	RCVD_IN_RP_SAFE	Sender is in Return Path Safe (trusted relay)
-0.001	SPF_HELO_NONE	SPF: HELO does not publish an SPF Record
0.001	SPF_PASS	SPF: sender matches SPF record **Great! Your SPF is valid**
-0.001	TVD_SPACE_RATIO	Uncommon space ratio

You are not allowed to use one of your sender email addresses -3

We check if the server you are sending from is authenticated

- [SPF] nikuska.cz does not allow your server 77.75.78.89 to use nikinka@nikuska.cz -3
- Your DKIM signature is valid
- Your message passed the DMARC test
- Your server **77.75.78.89** is successfully associated with **mxb.seznam.cz**
- Your domain name **nikuska.cz** is assigned to a mail server.
- Your hostname **mxb.seznam.cz** is assigned to a server.

Your message could be improved

You're not blocklisted

No broken links

Your lovely total: 0/10

Fake email with spam detected text:

Click here to view your message ✓

SpamAssassin does not like you -1000.2

The famous spam filter SpamAssassin. Score: -1000.2.
A score below -5 is considered spam.

-1000	GTUBE	Generic Test for Unsolicited Bulk Email
-1.985	PYZOR_CHECK	Similar message reported on Pyzor (https://www.pyzor.org) https://pyzor.readthedocs.io/en/latest/ **Please test a real content, test Newsletters will always be flagged by Pyzor** **Adjust your message or request whitelisting (https://www.pyzor.org)**
2	RCVD_IN_RP_SAFE	Sender is in Return Path Safe (trusted relay)
0.001	SPF_HELO_PASS	SPF: HELO matches SPF record
-0.001	SPF_NONE	SPF: sender does not publish an SPF Record
-0.001	TVD_SPACE_RATIO	Uncommon space ratio
-0.251	TVD_SPACE_RATIO_MINFP	Space ratio (vertical text obfuscation?)

You're not fully authenticated -2

We check if the server you are sending from is authenticated

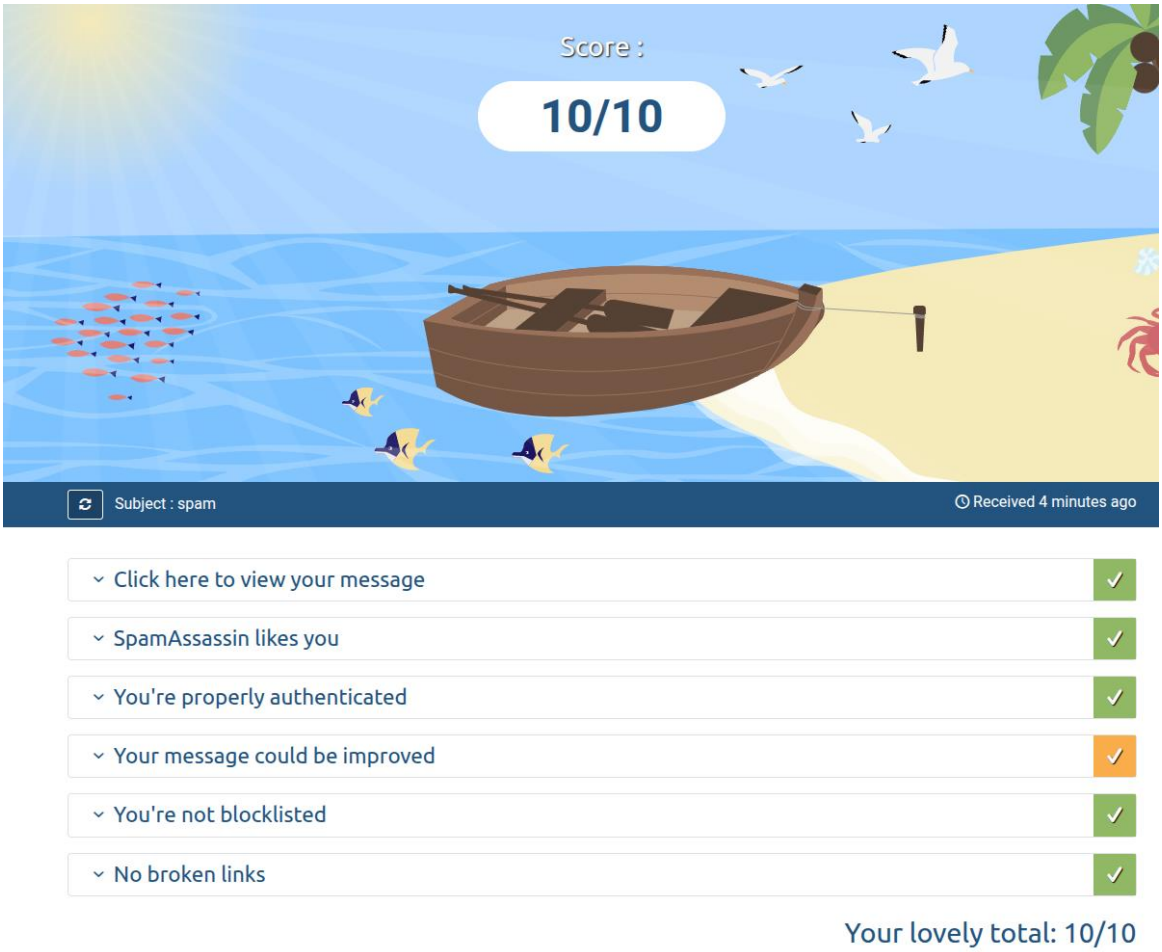
- [SPF] spam.com does not allow your server 114.29.236.247 to use text@spam.com -1
- Your message is not signed with DKIM -1
- You do not have a DMARC record ✓
- Your server **114.29.236.247** is successfully associated with **emkei.cz** ✓
- Your domain name **spam.com** is assigned to a mail server. ✓
- Your hostname **emkei.cz** is assigned to a server. ✓

Your message could be improved ✓

You're not blocklisted ✓

Your lovely total: 0/10

Report with look like spam text sent from a personal email:



Report with DNS correct setup sent with look like spam text:

- Click here to view your message ✓
- SpamAssassin likes you ✓
- You're properly authenticated ✓

We check if the server you are sending from is authenticated

- [SPF] Your server **77.75.78.89** is authorized to use **nikinka@nikuska.cz** ✓
- Your DKIM signature is valid ✓
- Your message passed the DMARC test ✓
- Your server **77.75.78.89** is successfully associated with **mxb.seznam.cz** ✓
- Your domain name **nikuska.cz** is assigned to a mail server. ✓
- Your hostname **mxb.seznam.cz** is assigned to a server. ✓

- Your message could be improved ✓
- You're not blocklisted ✓
- No broken links ✓

Your lovely total: 10/10

Report from mail sent from a DNS incorrect setup and look like spam text:

- Click here to view your message ✓
- SpamAssassin likes you ✓
- You are not allowed to use one of your sender email addresses -3

  We check if the server you are sending from is authenticated

  - [SPF] nikuska.cz does not allow your server 77.75.76.89 to use nikinka@nikuska.cz -3
  - Your DKIM signature is valid ✓
  - Your message passed the DMARC test ✓
  - Your server **77.75.76.89** is successfully associated with **mxb.seznam.cz** ✓
  - Your domain name **nikuska.cz** is assigned to a mail server. ✓
  - Your hostname **mxb.seznam.cz** is assigned to a server. ✓
- Your message could be improved ✓
- You're not blocklisted ✓
- No broken links ✓

Your lovely total: 7/10

Report of message sent from a fake mail with look like spam text:

Click here to view your message

SpamAssassin likes you

You're not fully authenticated -2

We check if the server you are sending from is authenticated

[SPF] spam.com does not allow your server 114.29.236.247 to use test@spam.com -1

Sender Policy Framework (SPF) is an email validation system designed to prevent email spam by detecting email spoofing, a common vulnerability, by verifying sender IP addresses.

You do not have a SPF record, please add the following one to your domain **spam.com**:

```
v=spf1 a mx ip4:114.29.236.247 ~all
```

The Name Server handling the domain name **spam.com** is **ns-457.awsdns-57.com.**

Need guidance? View guides for each host.

Verification details:

```
dig +short TXT spam.com :
  • "facebook-domain-verification=lmipv3urvsjn478zk72282o6tveiwj"
  • "MS=ms11310971"
  • "google-site-verification=DcQxVy7tQU1BmbwgqoXugoWML2hqDP4wBTCi35Gl3QI"
  • "ahrefs-site-verification_0290cc135d85fa21dbead34bff9e781913a52f73a91e11ed63b7529744464778"

dig +short +time=3 +tries=1 TXT @ns-457.awsdns-57.com. spam.com :
  • "MS=ms11310971"
  • "ahrefs-site-verification_0290cc135d85fa21dbead34bff9e781913a52f73a91e11ed63b7529744464778"
  • "facebook-domain-verification=lmipv3urvsjn478zk72282o6tveiwj"
  • "google-site-verification=DcQxVy7tQU1BmbwgqoXugoWML2hqDP4wBTCi35Gl3QI"

spfquery --scope mfrom --id test@spam.com --ip 114.29.236.247 --helo-id emkei.cz :
  • none
  • spam.com: No applicable sender policy available
  • spam.com: No applicable sender policy available
  • Received-SPF: none (spam.com: No applicable sender policy available) receiver=ns303428.ip-94-23-206.eu; identity=ma:
```

Your message is not signed with DKIM -1

You do not have a DMARC record

Your server **114.29.236.247** is successfully associated with **emkei.cz**

Your domain name **spam.com** is assigned to a mail server.

Your hostname **emkei.cz** is assigned to a server.

Your message could be improved

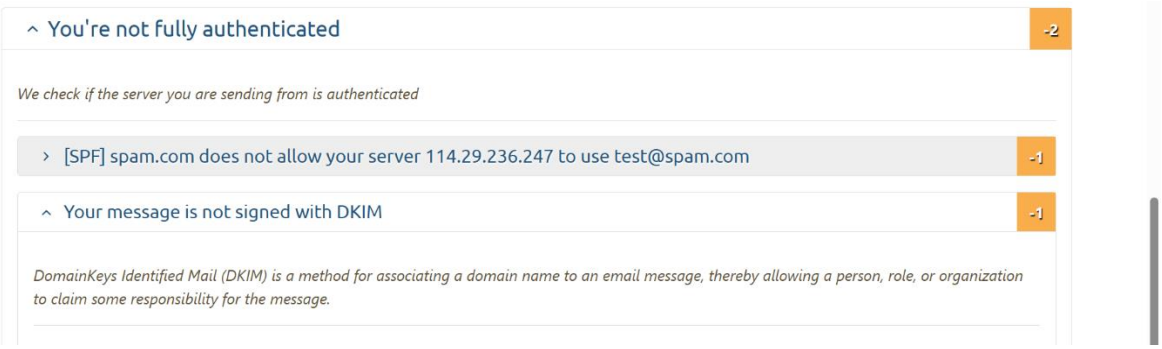
You're not blocklisted

Your lovely total: 8/10

Summarization of this experiment and achieved results

	Personal email	DNS setup correct	DNS setup incorrect	Fake email
Valid text	10	10	7	8
Spam detected text	0	0	0	0
Look like spam text	10	10	7	8

This table shows the results from Mail-tester after sending all 12 combinations of mail bodies and sending them from different emails with different setups. When sending from personal email with body of valid text or look like spam text I got score 10, this means everything is set up correctly and the spam text either does not look like spam, or the Mail-tester is not properly checking the content of the body email. With the DNS correct setup of my own domain I also got score of 10 when the body of an email is valid text or looks like spam. It get interesting when I changed the DNS setup to an incorrect one. Then the score dropped from 10 to 7. In the screenshot it is explicitly written that my domain does not allow to use the ipv4 address 77.75.76.89 which is one of the ipv4 addresses of seznam.cz so Mail-tester assigned 7 points to this setup. This happened because of setting "ipv4: 203.0.113.1", it could be fixed by setting "include: spf.seznam.cz" or setting "ipv4: 77.75.76.89". I also got score 7 with email body that looks like spam text. I also tested sending fake email from a fake email address test@spam.com using online tool Emkei's email sender. After evaluating the validity of the sent emails (valid text and look like spam) Mail-tester assigned 8 points to this setup. I got -1 point for not allowing SPF to use 114.29.236.247 and also -1 point for not signing the email with DKIM. Since it is not possible to set SPF or DKIM in Emkei's mail sender I cannot do anything about the score.



Lastly, using a specific known spam text inside the text field in the body gets immediately marked as spam and gets score 0, it is marked by SpamAssassin, no matter the DNS setup or no matter from which email address I used to send the email.

Valid text means sending an email with this body text:

Hi,

how are you? Long time no see.

Nikola

Spam detected text means sending an email with this text body:

*XJS*C4JDBQADN1.NSBN3*2IDNEN*GTUBE-STANDARD-ANTI-UBE-TEST-EMAIL*C.34X*

Look like spam text means sending an email with this body:

Hello mister,

I would like to report a virus on your computer. You probably clicked on a link in a spam email.

Reply to:

spam@spamcenter.com

Personal email means sending a message from: nika.sasinkova@gmail.com

DNS setup correct means sending an email from: nikinka@nikuska.cz with this DNS setup:

@	TXT	v=spf1 include:spf.seznam.cz ~all	1800	
_dmarc	TXT	v=DMARC1; p=reject;	1800	
szn20221014._domainkey	CNAME	szn20221014._domainkey.seznam.cz.	1800	

This SPF setting allows mail sent from Seznam.cz's servers and applies a soft fail to other sources (~all) that means many receivers will not outright reject a mail, but may mark it as suspicious, which means the email still should be delivered. This DMARC setup rejects unauthenticated emails other functionalities such as DKIM/SPF alignment settings or reporting

options are not set. DKIM signing is handled by Seznam's mail servers. This record links nikuska.cz to Seznam's DKIM public key.

DNS setup incorrect means sending an email from: nikinka@nikuska.cz with this DNS setup:

@	TXT	v=spf1 ip4:203.0.113.1 -all	1800
_dmarc	TXT	v=DMARC1; p=reject; pct=100; adkim=r; aspf=r	1800
szn20221014._domainkey	CNAME	szn20221014._domainkey.seznam.cz.	1800

This SPF setup allows only the IPv4 address 203.0.113.1 to send mail for your domain, with a hard fail for all others (-all). DMARC setting instructs receivers to reject non-authenticated emails, applies the policy to 100% of mail and checks both DKIM and SPF alignment in relaxed mode. The relaxed mode means that 2 addresses are aligned even if the names do not match, just domains have to, eg. nika.sasinkova@gmail.com and nikola.sasinkova@gmail.com . DKIM signing is handled by Seznam's mail servers. This record links nikuska.cz to Seznam's DKIM public key.

Fake email means sending a message from test@spam.com

Email parsing python script - my own original solution

In this experiment I used python 3.12 and PyCharm and python libraries like such as email, argparse, json and typing. Email library has a class ByteParser that lets you read an .eml email as bytes. With this I created a function parse_email_header() that uses ByteParser to read .eml file.

Then I extract header keywords I wanted to extract (I listed them in TARGET_HEADERS) and stored everything in dictionary with key being the header keyword and value being the data from the email header. I used library typing to ensure that function parse_email_header() returns a dictionary. In the main function I created a functionality to run the code from cmd with parameter file_path that must specify the path of a .eml file I want to parse. After this I call the function parse_email_header() and print the result as json so it prints in a nice structure.

Experiment with MXToolBox

The experimental infrastructure consists of my domain nikuska.cz, which is managed in the provider Email Profi DNS system. For the purposes of the experiment, SPF, DKIM and DMARC records are configured, but I will test SPF and DMARC incorrect configuration. The configuration is tested using MXToolbox tools. The results of these tests are recorded and their analysis is used to verify the correctness of the mail infrastructure settings.

The second test I am going to perform and describe is using MXToolBox for identifying issues with DNS records. This will be done by using section “Email Health” and pasting my domain name - nikuska.cz, then letting the MXToolBox diagnose my email health. For this part I will also use the correct and incorrect DNS setup used in previous tests:

Correct

@	TXT	v=spf1 include:spf.seznam.cz ~all	1800
_dmarc	TXT	v=DMARC1; p=reject;	1800
szn20221014._domainkey	CNAME	szn20221014._domainkey.seznam.cz.	1800

Incorrect SPF

@	TXT	v=spf1 ip4:203.0.113.1 -all	1800
_dmarc	TXT	v=DMARC1; p=reject; pct=100; adkim=r; aspf=r	1800
szn20221014._domainkey	CNAME	szn20221014._domainkey.seznam.cz.	1800

Incorrect DMARC

@	TXT	v=spf1 include:spf.seznam.cz ~all	1800
_dmarc	TXT	v=DMARC1; p=rejct; pct=100; adkim=r; aspf=r	1800
szn20221014._domainkey	CNAME	szn20221014._domainkey.seznam.cz.	1800

Reports from MXToolBox with incorrect SPF

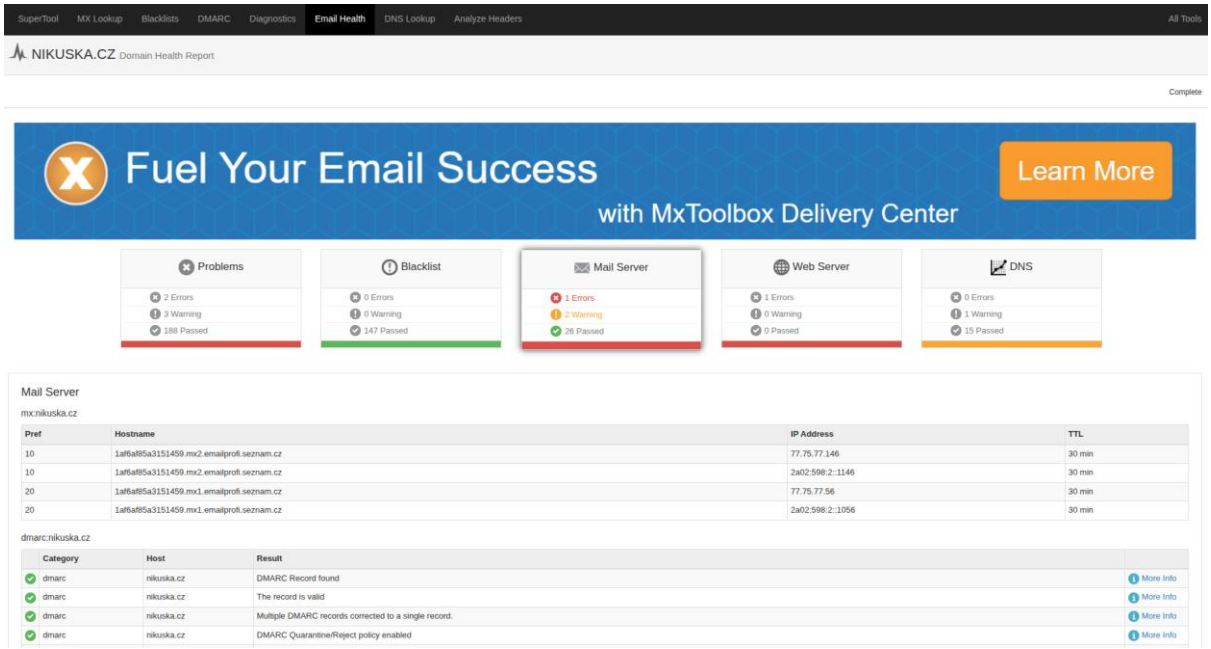
Here I incorrectly set SPF and correctly DKIM and DMARC:

DNS záznamy

Doména	Typ	Data	TTL
@	SOA	ans.emailprofi.cz. hostmaster.seznam.cz. 44 14400 3600 1814400 3600	3600
	MX	10 1af6af85a3151459.mx2.emailprofi.seznam.cz.	1800
	MX	20 1af6af85a3151459.mx1.emailprofi.seznam.cz.	1800
	NS	ams.emailprofi.cz.	1800
	NS	ans.emailprofi.cz.	1800
@	TXT	v=spf1 ipv4:203.0.113.1 -all	1800
_dmarc	TXT	v=DMARC1; p=reject; pct=100; adkim=r; aspf=r	1800
szn20221014._domainkey	CNAME	szn20221014._domainkey.seznam.cz.	1800

+ Vytvořit nový DNS záznam

Mail Server report from MXToolBox



I am going to focus on the section Mail Server since I am setting DNS records for emails. Here we see an error in Mail Server section.

Mail Server

mx:nikuska.cz

Pref	Hostname	IP Address	TTL
10	1af6af85a3151459.mx2.emailprofi.seznam.cz	77.75.77.146	30 min
10	1af6af85a3151459.mx2.emailprofi.seznam.cz	2a02:598:2::1146	30 min
20	1af6af85a3151459.mx1.emailprofi.seznam.cz	77.75.77.56	30 min
20	1af6af85a3151459.mx1.emailprofi.seznam.cz	2a02:598:2::1056	30 min

dmarc:nikuska.cz

	Category	Host	Result	
✓	dmarc	nikuska.cz	DMARC Record found	More Info
✓	dmarc	nikuska.cz	The record is valid	More Info
✓	dmarc	nikuska.cz	Multiple DMARC records corrected to a single record.	More Info
✓	dmarc	nikuska.cz	DMARC Quarantine/Reject policy enabled	More Info
✓	dmarc	nikuska.cz	All external domains in your DMARC record are giving permission to send them DMARC reports.	More Info

smtp:1af6af85a3151459.mx2.emailprofi.seznam.cz

	Category	Host	Result	
!	smtp	1af6af85a3151459.mx2.emailprofi.seznam.cz	Reverse DNS does not match SMTP Banner	More Info
✓	smtp	1af6af85a3151459.mx2.emailprofi.seznam.cz	OK - 77.75.77.146 resolves to wildcard.mx2.emailprofi.seznam.cz	More Info
✓	smtp	1af6af85a3151459.mx2.emailprofi.seznam.cz	OK - Reverse DNS is a valid Hostname	More Info
✓	smtp	1af6af85a3151459.mx2.emailprofi.seznam.cz	OK - Supports TLS.	More Info
✓	smtp	1af6af85a3151459.mx2.emailprofi.seznam.cz	1.389 seconds - Good on Connection time	More Info
✓	smtp	1af6af85a3151459.mx2.emailprofi.seznam.cz	OK - Not an open relay.	More Info
✓	smtp	1af6af85a3151459.mx2.emailprofi.seznam.cz	2.249 seconds - Good on Transaction Time	More Info

There is a warning about reverse DNS not matching SMTP Banner, it means email server sends a message called an SMTP banner when another server connects to it on port 25. This banner usually shows the server's name and any information the admin wants to share. The best practice for this banner is to include the server's actual hostname. This helps anyone connecting to know who they are talking to. I see a warning if the hostname in your SMTP banner doesn't match the hostname that's found by doing a reverse DNS lookup (PTR record) on my server's IP address. [22]

Session Transcript:

```
Transcript: Connecting to 77.75.77.146
Transcript:
Transcript: 220 email.seznam.cz ESMTP [1264 ms]
Transcript: EHLO keeper-us-east-1d.mxtoolbox.com
Transcript: 250-email.seznam.cz email zdarma na cely zivot (session-id:7a7345e8-eaeb-4d64-a8cc-dcd62eff61cf)
Transcript: 250-8BITMIME
Transcript: 250-PIPELINING
Transcript: 250-SIZE 52428800
Transcript: 250-STARTTLS
Transcript: 250 ENHANCEDSTATUSCODES [234 ms]
Transcript: MAIL FROM:<supertool@mxtoolboxsmtpdiag.com>
Transcript: 250 2.1.0 ok [281 ms]
Transcript: RCPT TO:<test@mxtoolboxsmtpdiag.com>
Transcript: 550 5.1.1 sorry, no such mailbox here [234 ms]
Transcript:
Transcript:
Transcript: - LookupServer [smtp:1af6af85a3151459.mx2.emailprofi.seznam.cz] 3781ms
Transcript:
```

This is a SMTP diagnostic transcript, it shows:

- MXToolBox successfully connected to your domain's mail server
- The server supports standard SMTP features (8BITMIME, SIZE, STARTTLS)

- The server correctly rejects a nonexistent mailbox with a 550
- DNS lookup takes about 3781 ms [20]

smtp:1af6af85a3151459.mx1.emailprofi.seznam.cz

	Category	Host	Result	
!	smtp	1af6af85a3151459.mx1.emailprofi.seznam.cz	Reverse DNS does not match SMTP Banner	More Info
✓	smtp	1af6af85a3151459.mx1.emailprofi.seznam.cz	OK - 77.75.77.56 resolves to wildcard.mx1.emailprofi.seznam.cz	More Info
✓	smtp	1af6af85a3151459.mx1.emailprofi.seznam.cz	OK - Reverse DNS is a valid Hostname	More Info
✓	smtp	1af6af85a3151459.mx1.emailprofi.seznam.cz	OK - Supports TLS.	More Info
✓	smtp	1af6af85a3151459.mx1.emailprofi.seznam.cz	1.405 seconds - Good on Connection time	More Info
✓	smtp	1af6af85a3151459.mx1.emailprofi.seznam.cz	OK - Not an open relay.	More Info
✓	smtp	1af6af85a3151459.mx1.emailprofi.seznam.cz	2.280 seconds - Good on Transaction Time	More Info

Here is the same warning as above about SMTP banner.

Session Transcript:

```
Transcript: Connecting to 77.75.77.56
Transcript:
Transcript: 220 email.seznam.cz ESMTP [1280 ms]
Transcript: EHLO keeper-us-east-1d.mxtoolbox.com
Transcript: 250-email.seznam.cz email zdarma na cely zivot (session-id:cbd39f5b-b86c-4f9b-85fc-6c9310642efd)
Transcript: 250-8BITMIME
Transcript: 250-PIPELINING
Transcript: 250-SIZE 52428800
Transcript: 250-STARTTLS
Transcript: 250 ENHANCEDSTATUSCODES [234 ms]
Transcript: MAIL FROM:<supertool@mxtoolboxsmtpdiag.com>
Transcript: 250 2.1.0 ok [281 ms]
Transcript: RCPT TO:<test@mxtoolboxsmtpdiag.com>
Transcript: 550 5.1.1 sorry, no such mailbox here [250 ms]
Transcript:
Transcript:
Transcript: - LookupServer [smtp:1af6af85a3151459.mx1.emailprofi.seznam.cz] 3016ms
Transcript:
```

This transcript shows the properly functioning mail receiving, because it shows:

- Successful connection to mail server
- Server supports required SMTP features (8BITMIME, PIPELINING, STARTTLS)
- Envelope sender accepted
- Envelope recipient rejected correctly because of nonexistent mailbox
- Reverse lookup and DNS checks
- SMTP server is healthy and behaving as expected [20]

spf:nikuska.cz - 1 Tests Failed

	Category	Host	Result	
✗	spf	nikuska.cz	Invalid syntax found	More Info
✓	spf	nikuska.cz	SPF Record found	More Info
✓	spf	nikuska.cz	No deprecated records found	More Info
✓	spf	nikuska.cz	Less than two records found	More Info
✓	spf	nikuska.cz	No items after 'ALL'.	More Info
✓	spf	nikuska.cz	DMARC Record found	More Info
✓	spf	nikuska.cz	DMARC Quarantine/Reject policy enabled	More Info

Type	Domain Name	TTL	Record	Prefix	Value	PrefixDesc	Description
record					txt		v=spf1 ipv4:203.0.113.1 -all
v					spf1		The SPF record version
ipv4				+	203.0.113.1	Pass	Unknown
all				-		Fail	Always matches. It goes at the end of your record.

As I already mentioned there is an error in this section. It is a syntax error in my SPF record. The most common errors include:

- Mechanisms that perform DNS lookups (mx, a, ptr, exists, redirect, include) contain text rather than domains or hostnames
- Mechanisms contain a numerical value, when they require a domain or hostname
- Format of IP addresses for ip4 and ip6 mechanisms is incorrect

The issue is with the keyword ipv4, it is not correct, it should be ip4: ... [23]

DNS setup from MXToolBox

Dns Server

dns:nikuska.cz

	Category	Host	Result	
!	dns	nikuska.cz	SOA Serial Number Format is Invalid	More Info
✓	dns	nikuska.cz	DNS Record found	More Info
✓	dns	nikuska.cz	No Bad Glue Detected	More Info
✓	dns	nikuska.cz	At Least Two Name Servers Found	More Info
✓	dns	nikuska.cz	All name servers are responding	More Info
✓	dns	nikuska.cz	All of the name servers are Authoritative	More Info
✓	dns	nikuska.cz	Local NS list matches Parent NS list	More Info
✓	dns	nikuska.cz	Name Servers appear to be Dispersed	More Info
✓	dns	nikuska.cz	Name Servers have Public IP Addresses	More Info
✓	dns	nikuska.cz	Serial numbers match	More Info
✓	dns	nikuska.cz	Primary Name Server Listed At Parent	More Info
✓	dns	nikuska.cz	SOA Refresh Value is within the recommended range	More Info
✓	dns	nikuska.cz	SOA Retry Value is within the recommended range	More Info
✓	dns	nikuska.cz	SOA Expire Value within recommended limits	More Info
✓	dns	nikuska.cz	SOA Minimum TTL Value is within allowed values	More Info
✓	dns	nikuska.cz	No Open Recursive Name Server Detected	More Info

DNS

Type	Domain Name	IP Address	TTL	Time (ms)	Status	Auth	Parent	Local
NS	ams.emailprofi.cz	77.75.75.166	30 min	112	✓	✓	✓	✓
NS	ans.emailprofi.cz	77.75.74.238	30 min	117	✓	✓	✓	✓

The report above states the following:

- A DNS record for nikuska.cz exists
- At least two name servers are configured and both respond
- All name servers are authoritative
- The local NS list matches what the parent (.cz) zone says
- A primary name server is correctly listed

- SOA refresh, retry, expire and minimum TTL values are all within recommended ranges [20]

Session Transcript:

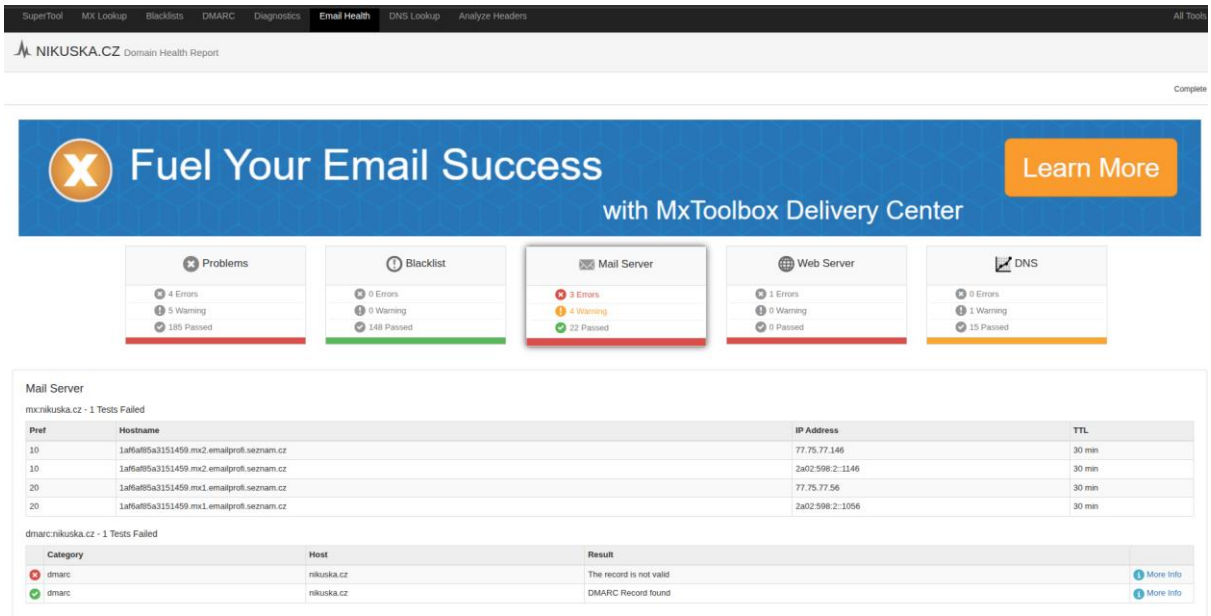
```
TimeStamp:

- LookupServer [dns:nikuska.cz] 2155ms

Depth: 1
ServerName: c.ns.nic.cz
ServerIP: 194.0.14.1
Authoritative: NON-AUTH
ElapsedTime: 140 ms
Result: Received 2 Referrals , rcode=NO_ERROR
Question:
Answers: nikuska.cz. 3600 IN NS ams.emailprofi.cz,nikuska.cz. 3600 IN NS ans.emailprofi.cz,

TimeStamp:
Depth: 2
ServerName: ams.emailprofi.cz
ServerIP: 77.75.75.166
Authoritative: AUTH
ElapsedTime: 125 ms
Result: Received 2 Answers , rcode=NO_ERROR
Question:
Answers: nikuska.cz. 1800 IN NS ams.emailprofi.cz,nikuska.cz. 1800 IN NS ans.emailprofi.cz,
```

This transcript shows how resolver walked the DNS tree to find records for my domain nikuska.cz. At first it asked c.ns.nic.cz which is one of the official .cz name servers, it returns 2 servers: ams.emailprofi.cz, ans.emailprofi.cz. Then it asked ams.emailprofi.cz for my domain. This was marked AUTH, because the server is authoritative for my domain. The whole process took 2155 ms. [20]

Incorrect DMARC setup

Here I incorrectly set DMARC and correctly SPF and DKIM:

Mail server report from MXToolBox



The first part states that all mail for my domain is handled by Seznam and IPv4 and IPv6 is present.

DMARC section states that 1 test failed:

- “DMARC Record found” - _dmarc.nikuska.cz exists.
- “The record is not valid” - The syntax of that TXT record is invalid because of p=“rject” instead of p=“reject”. [20]

smtp:1af6af85a3151459.mx2.emailprofi.seznam.cz

	Category	Host	Result	
!	smtp	1af6af85a3151459.mx2.emailprofi.seznam.cz	Reverse DNS does not match SMTP Banner	More Info
!	smtp	1af6af85a3151459.mx2.emailprofi.seznam.cz	May be an open relay.	More Info
✓	smtp	1af6af85a3151459.mx2.emailprofi.seznam.cz	OK - 77.75.77.146 resolves to wildcard.mx2.emailprofi.seznam.cz	More Info
✓	smtp	1af6af85a3151459.mx2.emailprofi.seznam.cz	OK - Reverse DNS is a valid Hostname	More Info
✓	smtp	1af6af85a3151459.mx2.emailprofi.seznam.cz	OK - Supports TLS.	More Info

Session Transcript:

```
Transcript: Connecting to 77.75.77.146
Transcript:
Transcript: 220 email.seznam.cz ESMTP [1177 ms]
Transcript: EHLO keeper-us-east-1d.mxtoolbox.com
Transcript: 250-email.seznam.cz email zdarma na cely zivot (session-id:6bde6970-0d6b-433c-ac02-11a2ffaf8499)
Transcript: 250-8BITMIME
Transcript: 250-PIPELINING
Transcript: 250-SIZE 52428800
Transcript: 250-STARTTLS
Transcript: 250 ENHANCEDSTATUSCODES [249 ms]
Transcript: MAIL FROM:<supertool@mxtoolboxsmtpdiag.com>
Transcript: 250 2.1.0 ok [263 ms]
Transcript: RCPT TO:<test@mxtoolboxsmtpdiag.com>
Transcript:
Transcript: SendSMTPCommand: You hung up on us after we connected. Please whitelist us. (connection lost)
Transcript:
Transcript:
Transcript: - LookupServer [smtp:1af6af85a3151459.mx2.emailprofi.seznam.cz] 2813ms
Transcript:
```

The first SMTP section states that there are 2 warnings:

- Reverse DNS does not match SMTP banner
- May be an open relay - The tool tried to send a test message from a random address to another random address, then decide if the server relays.

Everything else like: Reverse DNS, support TLS and ipv4 resolve are correct.

The session transcript shows:

- Support of 8BITMIME, PIPELINING, STARTTLS
- The server accepted the fake sender
- Connection drop from testing IP - Resulting in “may be an open relay” [20]

smtp:1af6af85a3151459.mx1.emailprofi.seznam.cz

	Category	Host	Result	
!	smtp	1af6af85a3151459.mx1.emailprofi.seznam.cz	Reverse DNS does not match SMTP Banner	More Info
!	smtp	1af6af85a3151459.mx1.emailprofi.seznam.cz	May be an open relay.	More Info
✓	smtp	1af6af85a3151459.mx1.emailprofi.seznam.cz	OK - 77.75.77.56 resolves to wildcard.mx1.emailprofi.seznam.cz	More Info
✓	smtp	1af6af85a3151459.mx1.emailprofi.seznam.cz	OK - Reverse DNS is a valid Hostname	More Info
✓	smtp	1af6af85a3151459.mx1.emailprofi.seznam.cz	OK - Supports TLS.	More Info

Session Transcript:

```
Transcript: Connecting to 77.75.77.56
Transcript:
Transcript: 220 email.seznam.cz ESMTP [1238 ms]
Transcript: EHLO keeper-us-east-1d.mxtoolbox.com
Transcript: 250-email.seznam.cz email zdarma na cely zivot (session-id:d8405b58-a460-40aa-85f3-c86c3167a1c7)
Transcript: 250-8BITMIME
Transcript: 250-PIPELINING
Transcript: 250-SIZE 52428800
Transcript: 250-STARTTLS
Transcript: 250 ENHANCEDSTATUSCODES [233 ms]
Transcript: MAIL FROM:<supertool@mxtoolboxsmtpdiag.com>
Transcript: 250 2.1.0 ok [287 ms]
Transcript: RCPT TO:<test@mxtoolboxsmtpdiag.com>
Transcript:
Transcript: SendSMTPCommand: You hung up on us after we connected. Please whitelist us. (connection lost)
Transcript:
Transcript:
Transcript: - LookupServer [smtp:1af6af85a3151459.mx1.emailprofi.seznam.cz] 2827ms
Transcript:
```

Same as the description above.

spf:nikuska.cz - 1 Tests Failed

	Category	Host	Result	
✗	spf	nikuska.cz	It is recommended to use a quarantine or reject policy. To enable BIMI, it is required to have one of these at 100%.	More Info
✓	spf	nikuska.cz	SPF Record found	More Info
✓	spf	nikuska.cz	No deprecated records found	More Info
✓	spf	nikuska.cz	Less than two records found	More Info
✓	spf	nikuska.cz	No items after 'ALL'.	More Info
✓	spf	nikuska.cz	The record is valid	More Info
✓	spf	nikuska.cz	Number of included lookups is OK	More Info
✓	spf	nikuska.cz	Nor Recursive Loops on Includes	More Info
✓	spf	nikuska.cz	No Duplicate Includes Found	More Info
✓	spf	nikuska.cz	No type PTR found	More Info
✓	spf	nikuska.cz	Number of void lookups is OK	More Info
✓	spf	nikuska.cz	Number of MX Resource Records is OK	More Info
✓	spf	nikuska.cz	No Null DNS Lookups found	More Info
✓	spf	nikuska.cz	DMARC Record found	More Info

Type	Domain Name	TTL	Record	Prefix	Value	PrefixDesc	Description
record					txt		v=spf1 include:spf.seznam.cz ~all
v					spf1		The SPF record version
include				+	spf.seznam.cz	Pass	The specified domain is searched for an 'allow'.
all				~		SoftFail	Always matches. It goes at the end of your record.

The first error is related to DMARC being wrong. It indicates that my domain is not protected against phishing and spoofing threats. To resolve this issue, I would need to set a Quarantine or Reject policy on the domain's DMARC record. [20]

	Category	Host	Result	
✗	http	nikuska.cz	The remote name could not be resolved: 'nikuska.cz' (http://nikuska.cz)	More Info

Lastly, I see an error about http: The remote name could not be resolved: 'nikuska.cz' (http://nikuska.cz). [20]

MXToolbox checks all kinds of pitfalls and verified all:

- SPF record found
- Syntax is valid
- No deprecated mechanisms, no loops, no excessive DNS lookups
- DMARC record exists

This results in SPF is correct and authorizes Seznam to send for nikuska.cz. [20]

DNS report from MXToolBox

Dns Server

dns:nikuska.cz

	Category	Host	Result	
!	dns	nikuska.cz	SOA Serial Number Format is Invalid	More Info
✓	dns	nikuska.cz	DNS Record found	More Info
✓	dns	nikuska.cz	No Bad Glue Detected	More Info
✓	dns	nikuska.cz	At Least Two Name Servers Found	More Info
✓	dns	nikuska.cz	All name servers are responding	More Info
✓	dns	nikuska.cz	All of the name servers are Authoritative	More Info
✓	dns	nikuska.cz	Local NS list matches Parent NS list	More Info
✓	dns	nikuska.cz	Name Servers appear to be Dispersed	More Info
✓	dns	nikuska.cz	Name Servers have Public IP Addresses	More Info
✓	dns	nikuska.cz	Serial numbers match	More Info
✓	dns	nikuska.cz	Primary Name Server Listed At Parent	More Info
✓	dns	nikuska.cz	SOA Refresh Value is within the recommended range	More Info
✓	dns	nikuska.cz	SOA Retry Value is within the recommended range	More Info
✓	dns	nikuska.cz	SOA Expire Value within recommended limits	More Info
✓	dns	nikuska.cz	SOA Minimum TTL Value is within allowed values	More Info
✓	dns	nikuska.cz	No Open Recursive Name Server Detected	More Info

DNS

Type	Domain Name	IP Address	TTL	Time (ms)	Status	Auth	Parent	Local
NS	ams.emailprofi.cz	77.75.75.166	30 min	116	✓	✓	✓	✓
NS	ans.emailprofi.cz	77.75.74.238	30 min	111	✓	✓	✓	✓

Everything is correct except 1 warning that states “SOA Serial Number Format is Invalid”, meaning SOA serial value is fine functionally, but its format does not follow the recommended style (usually YYYYMMDDnn, e.g. 2025112001). [20]

Session Transcript:

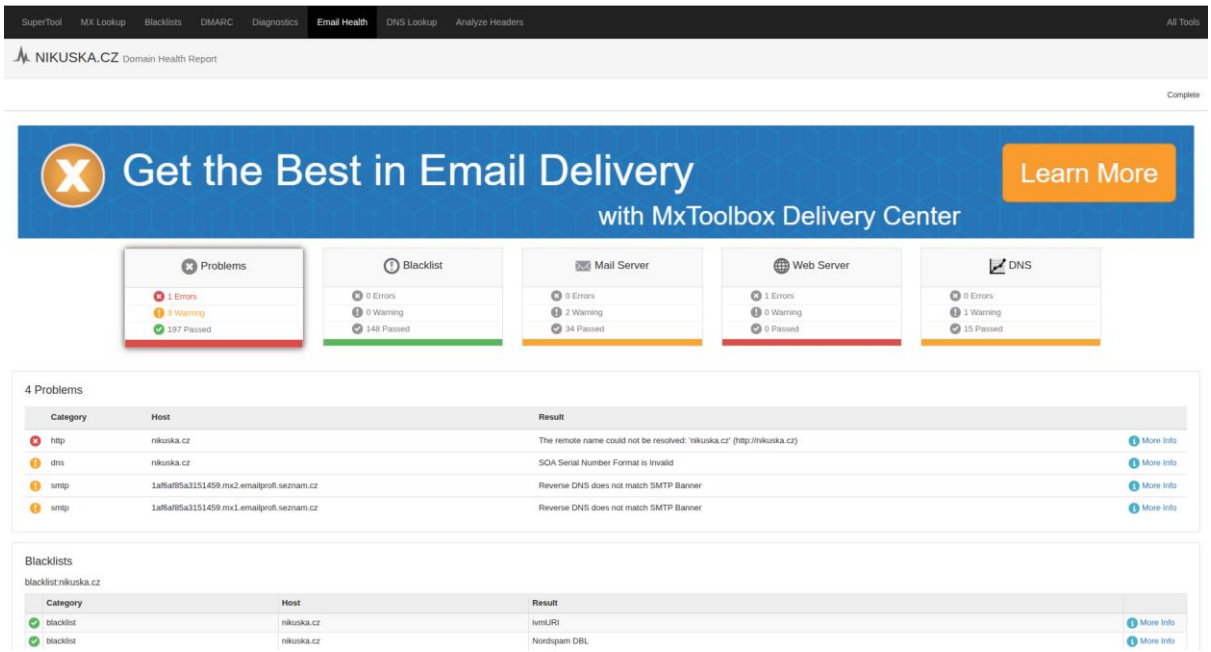
```
TimeStamp:

- LookupServer [dns:nikuska.cz] 2057ms

Depth: 1
ServerName: d.ns.nic.cz
ServerIP: 193.29.206.1
Authoritative: NON-AUTH
ElapsedTime: 101 ms
Result: Received 2 Referrals , rcode=NO_ERROR
Question:
Answers: nikuska.cz. 3600 IN NS ams.emailprofi.cz,nikuska.cz. 3600 IN NS ans.emailprofi.cz,

TimeStamp:
Depth: 2
ServerName: ans.emailprofi.cz
ServerIP: 77.75.74.238
Authoritative: AUTH
ElapsedTime: 111 ms
Result: Received 2 Answers , rcode=NO_ERROR
Question:
Answers: nikuska.cz. 1800 IN NS ans.emailprofi.cz,nikuska.cz. 1800 IN NS ams.emailprofi.cz,
```

This transcript shows how resolver walked the DNS tree to find records for my domain nikuska.cz. At first it asked d.ns.nic.cz which is one of the official .cz name servers, it returns 2 servers: ans.emailprofi.cz, ans.emailprofi.cz. Then it asked ans.emailprofi.cz for my domain. This was marked AUTH, because the server is authoritative for my domain. The whole process took 2057 ms. [20]

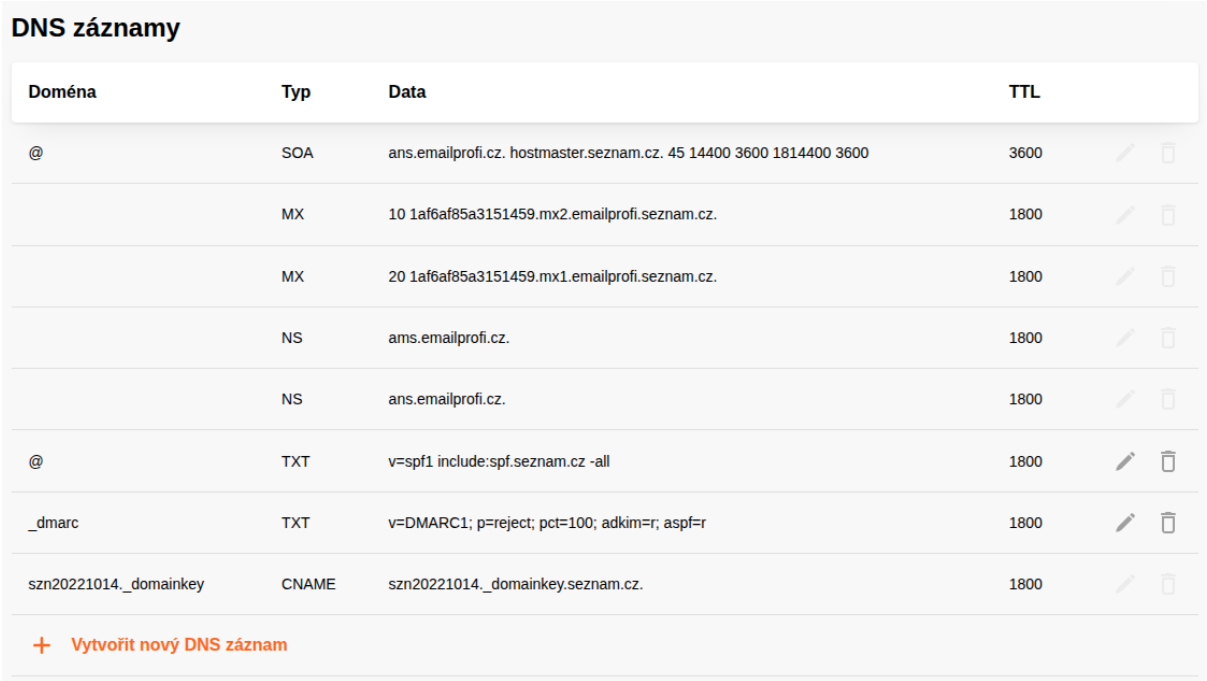
Correct DNS setup

Here I set everything (SPF, DMARC) correctly:

Mail Server report from MXToolBox



Now we do not see any errors in Mail Server section since everything is set up correctly.



DNS záznamy

Doména	Typ	Data	TTL
@	SOA	ans.emailprofi.cz. hostmaster.seznam.cz. 45 14400 3600 1814400 3600	3600
	MX	10 1af6af85a3151459.mx2.emailprofi.seznam.cz.	1800
	MX	20 1af6af85a3151459.mx1.emailprofi.seznam.cz.	1800
	NS	ams.emailprofi.cz.	1800
	NS	ans.emailprofi.cz.	1800
@	TXT	v=spf1 include:spf.seznam.cz -all	1800
_dmarc	TXT	v=DMARC1; p=reject; pct=100; adkim=r; aspf=r	1800
szn20221014._domainkey	CNAME	szn20221014._domainkey.seznam.cz.	1800

+ Vytvořit nový DNS záznam

Test results:

Mail Server

mx:nikuska.cz

Pref	Hostname	IP Address	TTL
10	1af6af85a3151459.mx2.emailprofi.seznam.cz	77.75.77.146	30 min
10	1af6af85a3151459.mx2.emailprofi.seznam.cz	2a02:598:2::1146	30 min
20	1af6af85a3151459.mx1.emailprofi.seznam.cz	77.75.77.56	30 min
20	1af6af85a3151459.mx1.emailprofi.seznam.cz	2a02:598:2::1056	30 min

dmarc:nikuska.cz

	Category	Host	Result	
✓	dmarc	nikuska.cz	DMARC Record found	More Info
✓	dmarc	nikuska.cz	The record is valid	More Info
✓	dmarc	nikuska.cz	Multiple DMARC records corrected to a single record.	More Info
✓	dmarc	nikuska.cz	DMARC Quarantine/Reject policy enabled	More Info
✓	dmarc	nikuska.cz	All external domains in your DMARC record are giving permission to send them DMARC reports.	More Info

smtp:1af6af85a3151459.mx2.emailprofi.seznam.cz

	Category	Host	Result	
!	smtp	1af6af85a3151459.mx2.emailprofi.seznam.cz	Reverse DNS does not match SMTP Banner	More Info
✓	smtp	1af6af85a3151459.mx2.emailprofi.seznam.cz	OK - 77.75.77.146 resolves to wildcard.mx2.emailprofi.seznam.cz	More Info
✓	smtp	1af6af85a3151459.mx2.emailprofi.seznam.cz	OK - Reverse DNS is a valid Hostname	More Info
✓	smtp	1af6af85a3151459.mx2.emailprofi.seznam.cz	OK - Supports TLS.	More Info
✓	smtp	1af6af85a3151459.mx2.emailprofi.seznam.cz	1.397 seconds - Good on Connection time	More Info
✓	smtp	1af6af85a3151459.mx2.emailprofi.seznam.cz	OK - Not an open relay.	More Info
✓	smtp	1af6af85a3151459.mx2.emailprofi.seznam.cz	2.329 seconds - Good on Transaction Time	More Info

The first part states that all mail for my domain is handled by Seznam and IPv4 and IPv6 is present. The second part states DMARC:

- That it is present
- It is correct
- Quarantine policy is enabled [20]

The last part states the SMTP check on MX server, it shows a warning – Reverse DNS does not match SMTP Banner, then everything else is OK:

- Reverse DNS is a valid Hostname
- Supports TLS,
- Good connection time
- Good transaction time [20]

Session Transcript:

```
Transcript: Connecting to 77.75.77.146
Transcript:
Transcript: 220 email.seznam.cz ESMTP [1281 ms]
Transcript: EHLO keeper-us-east-1d.mxtoolbox.com
Transcript: 250-email.seznam.cz email zdarma na cely zivot (session-id:a36baeb6-39a2-4d9e-ac93-deba1aa8e0c5)
Transcript: 250-8BITMIME
Transcript: 250-PIPELINING
Transcript: 250-SIZE 52428800
Transcript: 250-STARTTLS
Transcript: 250 ENHANCEDSTATUSCODES [240 ms]
Transcript: MAIL FROM:<supertool@mxtoolboxsmtpdiag.com>
Transcript: 250 2.1.0 ok [335 ms]
Transcript: RCPT TO:<test@mxtoolboxsmtpdiag.com>
Transcript: 550 5.1.1 sorry, no such mailbox here [244 ms]
Transcript:
Transcript:
Transcript: - LookupServer [smtp:1af6af85a3151459.mx2.emailprofi.seznam.cz] 3275ms
Transcript:
```

This is an SMTP diagnostic transcript, it shows:

- MXToolBox successfully connected to your domain's mail server
- The server supports standard SMTP features (8BITMIME, SIZE, STARTTLS)
- The server correctly rejects a nonexistent mailbox with a 550
- DNS lookup takes about 3275 ms [20]

smtp:1af6af85a3151459.mx1.emailprofi.seznam.cz

	Category	Host	Result	
!	smtp	1af6af85a3151459.mx1.emailprofi.seznam.cz	Reverse DNS does not match SMTP Banner	More Info
✓	smtp	1af6af85a3151459.mx1.emailprofi.seznam.cz	OK - 77.75.77.56 resolves to wildcard.mx1.emailprofi.seznam.cz	More Info
✓	smtp	1af6af85a3151459.mx1.emailprofi.seznam.cz	OK - Reverse DNS is a valid Hostname	More Info
✓	smtp	1af6af85a3151459.mx1.emailprofi.seznam.cz	OK - Supports TLS.	More Info
✓	smtp	1af6af85a3151459.mx1.emailprofi.seznam.cz	1.461 seconds - Good on Connection time	More Info
✓	smtp	1af6af85a3151459.mx1.emailprofi.seznam.cz	OK - Not an open relay.	More Info
✓	smtp	1af6af85a3151459.mx1.emailprofi.seznam.cz	2.413 seconds - Good on Transaction Time	More Info

Session Transcript:

```
Transcript: Connecting to 77.75.77.56
Transcript:
Transcript: 220 email.seznam.cz ESMTP [1307 ms]
Transcript: EHLO keeper-us-east-1d.mxtoolbox.com
Transcript: 250-email.seznam.cz email zdarma na cely zivot (session-id:b7d53a00-545c-4902-9573-93e114e0c092)
Transcript: 250-8BITMIME
Transcript: 250-PIPELINING
Transcript: 250-SIZE 52428800
Transcript: 250-STARTTLS
Transcript: 250 ENHANCEDSTATUSCODES [247 ms]
Transcript: MAIL FROM:<supertool@mxtoolboxsmtpdiag.com>
Transcript: 250 2.1.0 ok [327 ms]
Transcript: RCPT TO:<test@mxtoolboxsmtpdiag.com>
Transcript: 550 5.1.1 sorry, no such mailbox here [259 ms]
Transcript:
Transcript:
Transcript: - LookupServer [smtp:1af6af85a3151459.mx1.emailprofi.seznam.cz] 3431ms
Transcript:
```

This transcript shows the properly functioning mail receiving, because it shows:

- Successful connection to mail server

- Server supports required SMTP features (8BITMIME, PIPELINING, STARTTLS)
- Envelope sender accepted
- Envelope recipient rejected correctly because of nonexistent mailbox
- Reverse lookup and DNS checks
- SMTP server is healthy and behaving as expected [20]

spf:nikuska.cz

	Category	Host	Result	
✓	spf	nikuska.cz	SPF Record found	More Info
✓	spf	nikuska.cz	No deprecated records found	More Info
✓	spf	nikuska.cz	Less than two records found	More Info
✓	spf	nikuska.cz	No items after 'ALL'.	More Info
✓	spf	nikuska.cz	The record is valid	More Info
✓	spf	nikuska.cz	Number of included lookups is OK	More Info
✓	spf	nikuska.cz	Nor Recursive Loops on Includes	More Info
✓	spf	nikuska.cz	No Duplicate Includes Found	More Info
✓	spf	nikuska.cz	No type PTR found	More Info
✓	spf	nikuska.cz	Number of void lookups is OK	More Info
✓	spf	nikuska.cz	Number of MX Resource Records is OK	More Info
✓	spf	nikuska.cz	No Null DNS Lookups found	More Info
✓	spf	nikuska.cz	DMARC Record found	More Info
✓	spf	nikuska.cz	DMARC Quarantine/Reject policy enabled	More Info

Type	Domain Name	TTL	Record	Prefix	Value	PrefixDesc	Description
record					txt		v=spf1 include:spf.seznam.cz -all
v					spf1		The SPF record version
include				+	spf.seznam.cz	Pass	The specified domain is searched for an 'allow'.
all				-		Fail	Always matches. It goes at the end of your record.

This screenshot shows that everything regarding SPF is setup correctly, there are no errors and the functionality is 100%.

DNS report from MXToolBox

Dns Server

dns:nikuska.cz

	Category	Host	Result	
!	dns	nikuska.cz	SOA Serial Number Format is Invalid	More Info
✓	dns	nikuska.cz	DNS Record found	More Info
✓	dns	nikuska.cz	No Bad Glue Detected	More Info
✓	dns	nikuska.cz	At Least Two Name Servers Found	More Info
✓	dns	nikuska.cz	All name servers are responding	More Info
✓	dns	nikuska.cz	All of the name servers are Authoritative	More Info
✓	dns	nikuska.cz	Local NS list matches Parent NS list	More Info
✓	dns	nikuska.cz	Name Servers appear to be Dispersed	More Info
✓	dns	nikuska.cz	Name Servers have Public IP Addresses	More Info
✓	dns	nikuska.cz	Serial numbers match	More Info
✓	dns	nikuska.cz	Primary Name Server Listed At Parent	More Info
✓	dns	nikuska.cz	SOA Refresh Value is within the recommended range	More Info
✓	dns	nikuska.cz	SOA Retry Value is within the recommended range	More Info
✓	dns	nikuska.cz	SOA Expire Value within recommended limits	More Info
✓	dns	nikuska.cz	SOA Minimum TTL Value is within allowed values	More Info
✓	dns	nikuska.cz	No Open Recursive Name Server Detected	More Info

DNS

Type	Domain Name	IP Address	TTL	Time (ms)	Status	Auth	Parent	Local
NS	ams.emailprofi.cz	77.75.75.166	30 min	116	✓	✓	✓	✓
NS	ans.emailprofi.cz	77.75.74.238	30 min	123	✓	✓	✓	✓

Session Transcript:

```
TimeStamp:

- LookupServer [dns:nikuska.cz] 1975ms

Depth: 1
ServerName: d.ns.nic.cz
ServerIP: 193.29.206.1
Authoritative: NON-AUTH
ElapsedTime: 94 ms
Result: Received 2 Referrals , rcode=NO_ERROR
Question:
Answers: nikuska.cz. 3600 IN NS ams.emailprofi.cz,nikuska.cz. 3600 IN NS ans.emailprofi.cz,

TimeStamp:
Depth: 2
ServerName: ans.emailprofi.cz
ServerIP: 77.75.74.238
Authoritative: AUTH
ElapsedTime: 122 ms
Result: Received 2 Answers , rcode=NO_ERROR
Question:
Answers: nikuska.cz. 1800 IN NS ans.emailprofi.cz,nikuska.cz. 1800 IN NS ams.emailprofi.cz,
```

This transcript shows how a resolver walked the DNS tree to find records for my domain nikuska.cz. At first it asked d.ns.nic.cz which is one of the official .cz name servers, it returns 2 servers: ams.emailprofi.cz, ans.emailprofi.cz. Then it asked ans.emailprofi.cz for my domain. This was marked AUTH, because the server is authoritative for my domain. The whole process took 1975 ms. [20]

Summarization of this experiment and achieved results

This experiment provided reports from MXToolBox and a comparison between correct DNS setup, incorrect SPF setup and incorrect DMARC setup. Below is the table comparing Mail server health of these setups:

	Errors	Warnings	Correct
DNS correct setup	0	2	34
SPF incorrect	1	2	26
DMARC incorrect	3	4	22

We can see the difference between the setups by the number of errors, warnings and correct passes.

When the setup was correct, there were just 2 warnings:

- Both are about SMTP that reverse DNS does not contain SMTP banner

With SPF setup incorrectly, we see an error:

- SPF - Invalid syntax found

Warnings:

- 2 are about reverse DNS does not match SMTP Banner

After setting up just incorrect DMARC we see more errors:

- SPF - Recommended to use quarantine or reject policy
- DMARC record is invalid
- (http could not be resolved) - Not important during this experiment

Then there are 4 warnings:

- 2 reverse DNS does not match SMTP Banner
- 2 may be an open relay

Below is a table comparing DNS reports from Mail health:

	Errors	Warnings	Correct
DNS correct setup	0	1	15
SPF incorrect	0	1	15
DMARC incorrect	0	1	15

During this experiment the setup did not have an effect on the results, there is exactly 1 waning that is the same for all setups:

- SOA Serial Number Format is invalid

Spam and Spam filters

Spam is any unwanted message that is sent to many people at the same time, usually without their permission. Most spam comes through email, but it can also appear in text messages, social media or online comments. Spam messages often try to sell products, promote fake offers or convince someone to click a link. Some spam might simply be annoying, but other spam can be dangerous because it may contain phishing links or harmful attachments. These could lead to malware infections, identity theft or the loss of personal data. Spammers usually send out large numbers of messages because even if only a small number of people respond, they can still make money or cause harm. Spam continues to grow and change as attackers find new ways to reach people, so understanding what spam is and why it is risky is an important part of maintaining good email security. By knowing the common signs of spam (unexpected messages, strange requests, poor grammar...) users can protect themselves and avoid becoming victims of cyber crime. [24]

Spam detection

Spam detection involves identifying patterns or features that are common in unwanted or harmful messages. One method is to examine the content of an email, such as suspicious keywords (free, urgent, limited offer...) or unusual formatting. Another method looks at the sender's information, such as whether the email address or IP address is known for sending harmful messages. Many systems use blacklists, which are databases of email addresses, domains or servers that have previously been identified as sources of spam. Some spam detection systems also examine the structure of the email (headers, language, attachments, links...) to decide if it looks irregular. Modern spam detection often uses machine learning. These systems learn from large numbers of examples and adapt to new types of spam. Because attackers constantly change their techniques, spam detection must also evolve. This means updating detection rules, analyzing new threats and comparing behavior patterns. Reliable spam detection helps reduce the number of unwanted messages and protects users from dangerous content. [24]

Spam filters

Spam filters use a combination of different techniques to stop unwanted or harmful emails from reaching the inbox. First, content filters examine the words, phrases and structure of the email to see if it matches known spam patterns. Header filters review information like the sender's address, server path and technical details to detect suspicious behavior. Blacklist filters block messages from email addresses, domains or IP addresses already known for sending spam. Rule-based filters apply custom rules that administrators or security teams create, such as blocking emails with certain attachments or specific keywords. More advanced filters use machine learning or Bayesian methods, which allow the system to improve over time by learning which emails users mark as spam. Spam filters usually score each message based on different checks and if the score is too high, the email is moved to a spam folder or blocked completely. Because spammers constantly change strategies - such as masking their identity or using new domains - effective spam filters must be updated regularly and rely on current threat intelligence. By combining multiple techniques, spam filters help keep inboxes clean, reduce security risks and protect organizations from phishing, malware and unwanted email traffic. [24]

Fortinet Antispam

Fortinet Antispam is a security service used in FortiMail and other Fortinet products to detect and block unwanted or dangerous emails before they reach the user. It combines several advanced techniques, such as IP reputation checking, URL filtering, content scanning and machine-learning analysis, to identify spam, phishing emails and messages that contain harmful links or attachments. Fortinet Antispam uses constantly updated threat intelligence from the global FortiGuard network, which tracks spam sources and new attack methods around the world. This allows the system to react quickly to new spam campaigns and improve accuracy over time. When Fortinet Antispam identifies a suspicious email, it can block it, quarantine it or simply tag it, depending on the security policy. Overall, Fortinet Antispam helps organizations keep their inboxes safe, reduce the risk of cyberattacks and maintain cleaner email traffic. [25] [26]

IP Reputation

One of the features of FortiGuard Antispam is the IP reputation check. When a new email arrives, FortiMail takes the sender's IP address (or, if configured, extracts public IPs from the "Received" headers) and sends it to the FortiGuard Antispam service. FortiGuard keeps a list of known bad or spammy IP addresses and it returns a reputation "level" so FortiMail can decide what to do. Based on this reputation, FortiMail applies a policy: very bad IPs might be rejected immediately, while less bad ones may be quarantined or just tagged. This is useful because many spam emails come from IP addresses that are already known to be problematic. By using reputation-based blocking, FortiMail reduces the chance of letting in spam from repeatedly abusive sources. [26]

URL Filtering

Another strong part of FortiGuard Antispam is URL filtering. In this method, FortiMail inspects the body of an email, finds all links (URLs) and checks them against a FortiGuard-maintained URL blacklist. The administrator can choose which categories of URLs to block (for example, hacking, malware or spam). If the URL belongs to a blocked category, FortiMail treats the email as spam and applies a defined action (such as quarantine, reject or tag). This is very powerful because many spam or phishing emails contain malicious links. Even if the text of the email looks safe, the URLs might lead to bad sites - so this check protects users from dangerous destinations. [26]

Spam Outbreak Protection

FortiGuard Antispam also has a spam outbreak protection feature. This is quite clever: if FortiMail sees many suspicious emails coming from unknown sources, it does not immediately reject them all - instead, it can temporarily hold them in a queue. During that holding time, FortiGuard may re-check its lists because maybe new information has appeared (for example a previously unknown spam server was added to blocklists). After a configured outbreak protection period, FortiMail re-evaluates the emails and then applies the right action (reject, quarantine, tag) depending on the result. This is technically useful because it avoids

false positives in fast-changing spam campaigns: rather than rejecting everything immediately, FortiMail gives FortiGuard time to catch up, improving both security and reliability. [26]

Heuristic and Rule-Based Scanning

FortiMail also supports heuristic scanning, which means it uses a set of rules to score an email. Each heuristic rule looks for certain patterns - like suspicious phrases, weird header behaviors or typical spam structures - and assigns a score to each rule if it matches. The total score of all matching rules is compared to a threshold: if it is above that threshold, FortiMail considers the email spam. Administrators can configure how many of the heuristic rules to use (percentage of rules used), which gives a good balance between detection power and performance. Because heuristic scanning is resource-intensive, FortiMail allows tuning (or disabling it for some profiles) to avoid over-loading the system. This method is interesting because it does not rely solely on blocklists because it helps detect new or unknown spam based on behaviour and content patterns. [26]

Bayesian Filtering

FortiGuard Antispam does not stop at blocklists and heuristics: it also supports Bayesian filtering. In FortiMail you can enable Bayesian scans in antispam profiles. Bayesian filtering works by analyzing previous emails that were either “spam” or “not spam” and using that history to calculate the probability that a new message is spam. FortiMail can use both a global Bayesian database and per-domain Bayesian databases, depending on how you configure it. There is also an option to train the Bayesian database: users can forward training messages (spam or good) to a control account so that FortiMail learns which patterns are really spammy. FortiMail can use results from other filters to train the Bayesian engine automatically - this helps the filter adapt more quickly. This is powerful because Bayesian filtering is adaptive: it learns over time, becoming more accurate based on real-world email traffic, reducing both false positives and missed spam. [26]

Image and PDF Spam Scanning

FortiMail's antispam profiles also offer image spam scanning. You can configure it to inspect embedded graphics (GIF, JPG, PNG) and even image attachments. If you enable "Aggressive" mode, FortiMail also scans images attached as files, not just images embedded in the email. In addition, FortiMail can scan the first page of PDF attachments: if a PDF does not have normal text, spammers may hide spam in that PDF. By scanning PDFs, FortiMail uses its heuristic rules, banned-word rules or image scanning to decide if the PDF is spammy. This is technically interesting because many spammers use "image-only" emails or PDFs to bypass simple text-based spam filters. By doing deeper scans of images and PDF content, FortiMail can catch those sneaky spam messages. [26]

Quarantine, Tag, Reject

Finally, FortiMail lets administrators decide exactly what to do when spam is detected, using antispam action profiles. According to the antispam profile configuration, you can tell FortiMail to reject the email, quarantine it (system or per-user) or tag the message (for example, add "[SPAM]" to the subject). These actions give flexibility: for less certain spam, you might want to quarantine and let a user review it; for very bad or high-risk spam, you may reject it immediately. Also, you can combine different actions depending on the scan type (e.g. FortiGuard, Bayesian, heuristic) because each scan type can have its own action profile. This gives strong control to administrators. They can tune the system to their needs: maximize strict blocking or lean toward caution (quarantine) to reduce the risk of blocking legitimate mail. [26]

Tips how to prevent spam mails and improve security

- Avoid unknown links and attachments: Do not blindly click any links or open attachments. Confirm the message by contacting the sender through a known and trusted methods.
- Flag spam messages: Use "mark as spam" or "report" feature. This helps improve spam filters for you and other users.

- Keep software up to date: Regularly update your email client, antivirus tools and operating system.
- Use Two-Factor authentication: Enable 2FA on your accounts to add an extra security layer. [27]

General tips on how to stay safe online

- Install up-to-date antivirus and firewall software: Have reputable security software and download updates regularly.
- Create a strong password: Mix of letters, symbols and numbers.
- Change password regularly: It is sensible to update your password from time to time
- Do not share passwords.
- Be alert to online scams: Be cautious of offers or deals that seem unusually generous or "too good to be true". [28]

Python Mail Sender - my own original solution

In this experiment I used python 3.12 and PyCharm and python libraries like email and smtplib, I authenticated myself to SMTP gmail.com by using App Password and generating a password for authentication, this is a 16 letters long password. This is used because sometimes it is hard to authenticate myself just by email and password.

The python program consists of 2 functions, 1st creates an email from arguments and sets SUBJECT, FROM, RECIPIENTS... and adds attachments one by one as bytes (application/octet-stream) into the email. The 2nd function sends already created email, it connects to the SMTP server, sets up encryption, authenticates me and then sends the message. In the main I can just setup the SENDER, SUBJECT, RECIPIENTS, BODY_TEXT, ATTACHMENTS and call 2 already mentioned functions to successfully send an email.

Comparison of MXToolBox and Mail-tester

Both MXToolBox and Mail-tester focus on email deliverability and its issues, but they focus on different layers. MXToolBox is primarily an infrastructure and diagnostics toolkit. It focuses on the technical configuration and health of your email environment: DNS, MX records, SPF, DKIM, DMARC, blacklists and SMTP server behaviour. With MXToolBox you can check whether your mail server is reachable, whether you are listed on common blacklists and whether your authentication records are set up correctly. In its paid plans it can also provide ongoing monitoring, alerts and more advanced deliverability and inbox placement analysis. This makes it particularly useful for system administrators or technically oriented users who are setting up a new domain, troubleshooting rejections or needing continuous visibility into mail flow and its reputation. Mail-tester, by contrast, focuses on the content and characteristics of a specific email message. The typical workflow is to send your newsletter or campaign to a special test address and then view a report that gives you a score out of ten, along with explanations. Mail-tester evaluates how spam filters might see that particular email. It checks whether SPF, DKIM and DMARC pass for the message, whether the sending IP or domain appears on blacklists and whether the content and HTML structure look suspicious. The interface is simple and oriented towards marketers or content creators who want to improve a campaign before sending it to real recipients. In practice, the tools are complementary rather than interchangeable. A sensible workflow is to use MXToolBox to ensure your technical setup and reputation are set up correctly and then use Mail-tester to fine-tune individual messages before major sends. [20] [16]

TLS

Transport Layer Security is the default encryption method used by most mainstream email services. When you send a message via mail providers, TLS creates an encrypted tunnel between mail servers so that anyone monitoring the network cannot easily read the contents in transit. [29]

In practice, your email client first negotiates a secure channel with your provider's server using a TLS handshake. During this process, the client and server authenticate each other, agree on cryptographic algorithms and establish shared keys. These keys then protect the data using fast symmetric encryption, while public-key cryptography is used mainly for the initial key exchange. [29]

TLS has an important limitation: it only protects messages while they move between servers. At each mail server, the message is briefly decrypted and then re-encrypted for the next hop. This means the provider can access email contents and messages are stored on servers in a readable form unless additional encryption is used. [29]

TLS is good enough, it shields you from passive eavesdropping, simple interception attempts and many man-in-the-middle attacks. But for highly sensitive information TLS alone is not sufficient, because it does not secure emails at rest or hide metadata such as sender, recipient and subject lines. [29]

E2EE

End-to-end encryption is designed to ensure that only you and your intended recipient can read an email's contents. Instead of merely encrypting data while it travels between servers, E2EE encrypts the message on the sender's device and keeps it encrypted until it is decrypted on the recipient's device. Service providers, network operators and other intermediaries should never see the plaintext. [29]

E2EE relies on public key cryptography. Each user has a key pair, a public key that can be shared with others and a private key that must remain secret. When someone wants to send you a secure message, they use your public key to transform the plaintext into ciphertext. Only your private key can reverse this process, so even if a server storing the message is compromised, the attacker should not be able to read it without that key. [29]

E2EE is not without drawbacks. Routing metadata, such as sender and recipient addresses, usually remains visible so mail systems can deliver the message. Both parties also need compatible E2EE setups, which can make deployment harder. Despite these challenges, E2EE is widely used for protecting truly sensitive communications. [29]

Encryption of email messages

TLS encrypts email only while it's traveling between servers, so the mail provider can still read the message once it's stored. E2EE encrypts the message on your device and keeps it encrypted until it reaches the recipient's device, so even the provider can't see the content. TLS is easier and works automatically with almost all email services but offers weaker protection at rest. E2EE is stronger for sensitive data and privacy but requires compatible tools on both sides and can be less convenient to set up. [29]

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